Hands-on: RNA-seq in Omics-Studio

Exercises 1–5 by hand · Exercise 6 with OMAKASE

FastQC → STAR → featureCounts → CountQC → edgeR

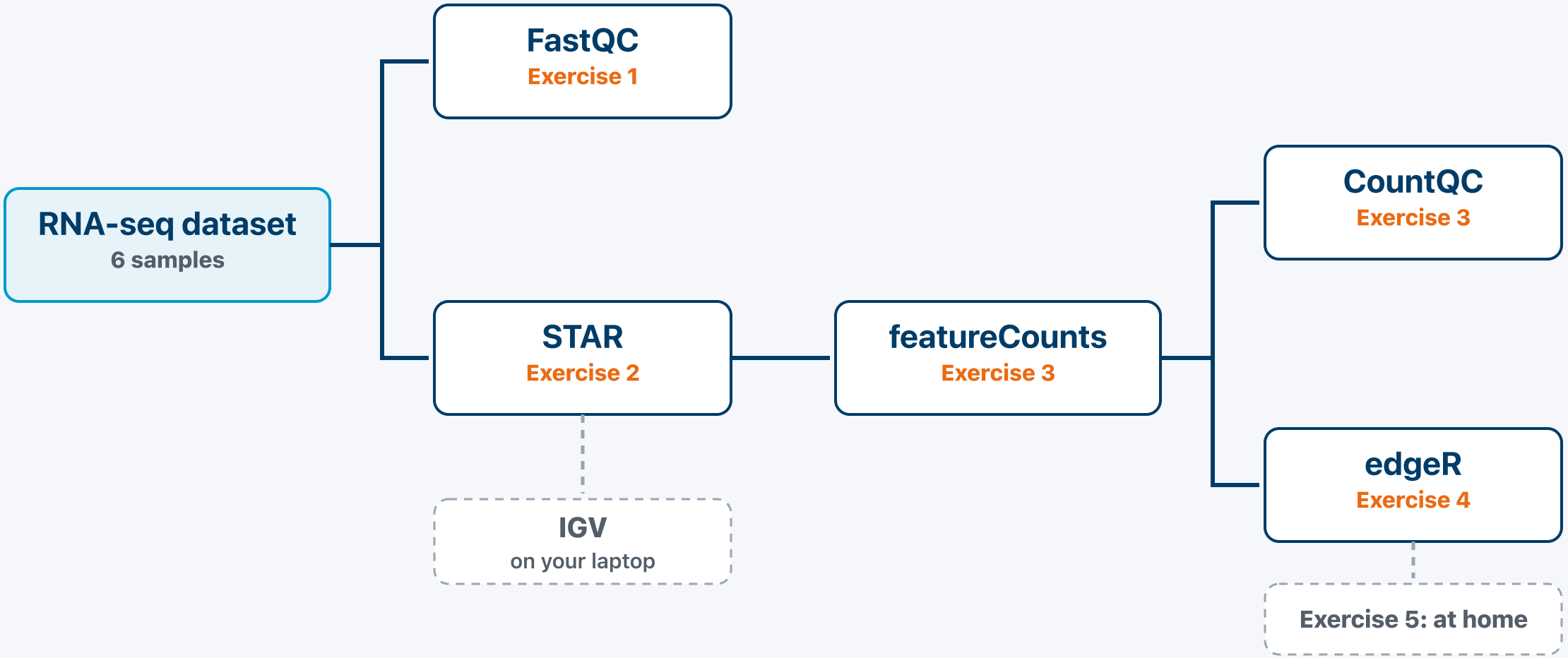
Bio610 · 28 October 2026 · 13.00–13.50 · Masaomi Hatakeyama · FGCZ

exercise pages

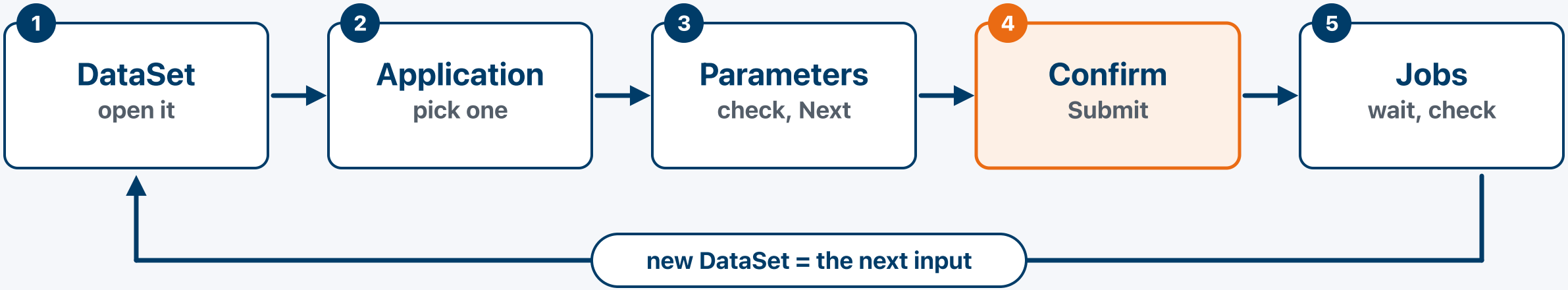
→ next · ↑ find a slide

Today you grow this dataset tree

box = dataset · line = one app run · parent = input, child = output



Every app: the same five screens



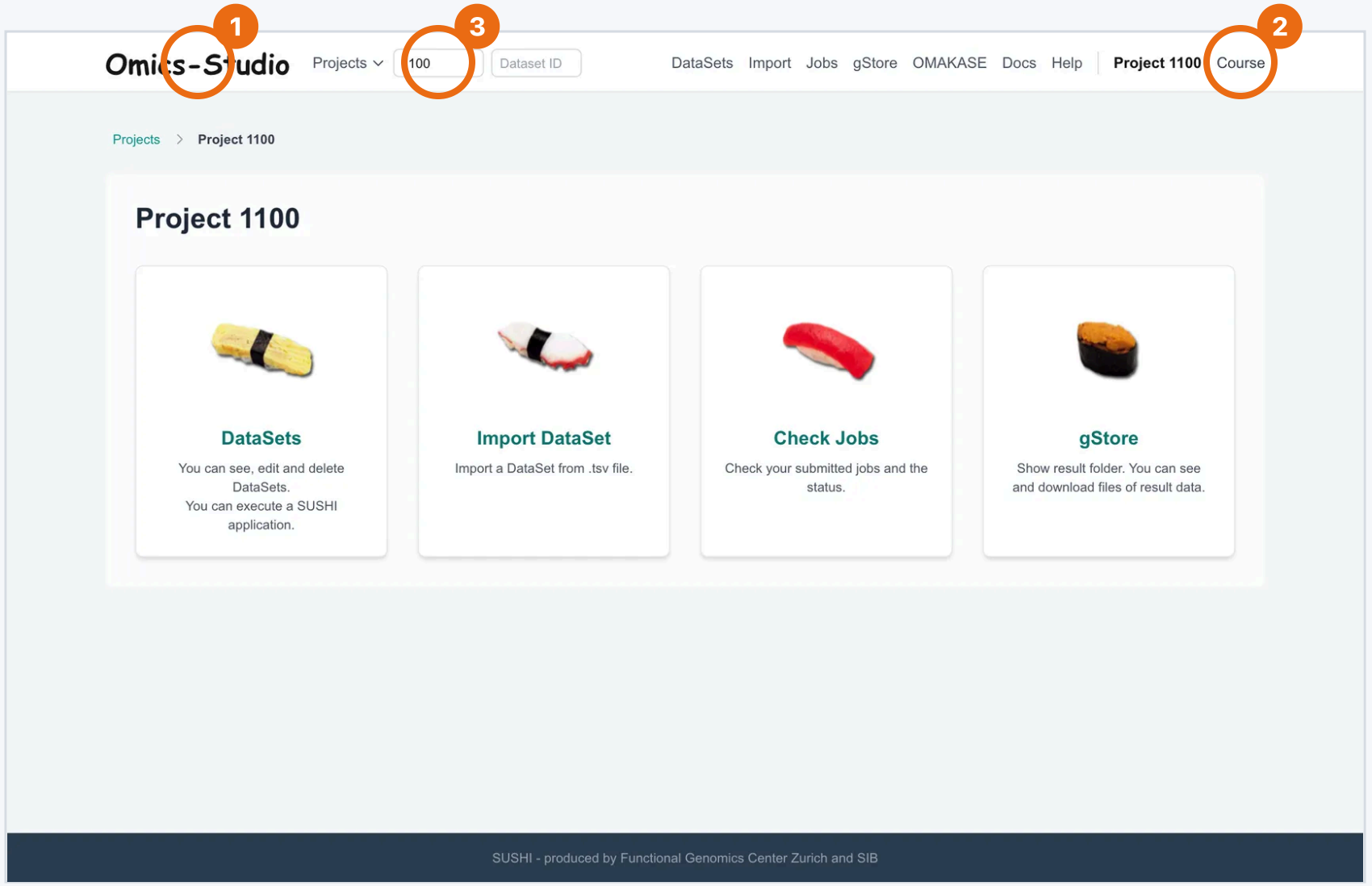
Same five screens

for FastQC, STAR, featureCounts, CountQC and edgeR

Where

course Omics-Studio · no sign-in
fgcz-course2.bfabric.org

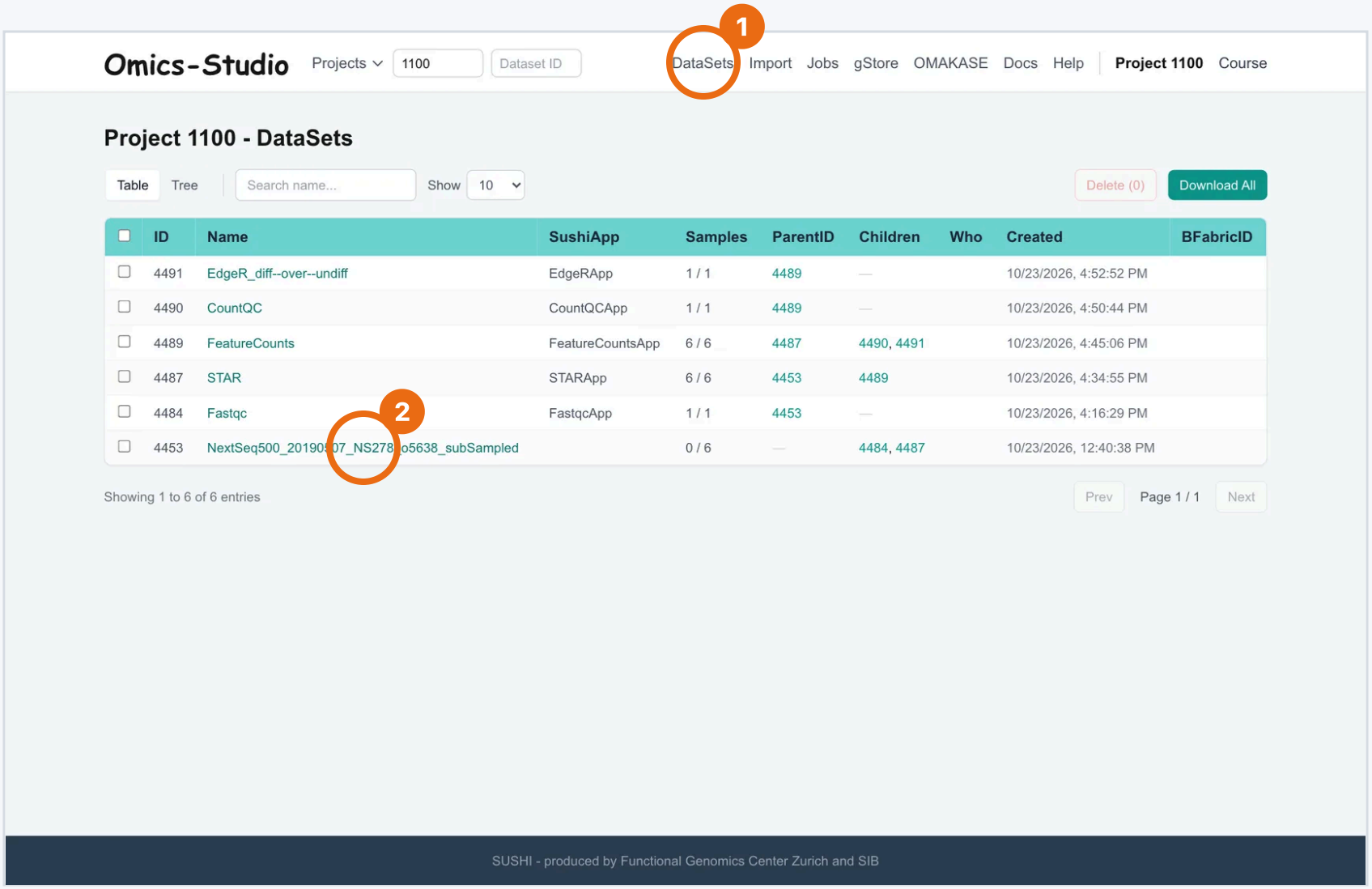
Open the course Omics-Studio



Start

1. **Omics-Studio: fgcz-course2.bfabric.org**
2. **Course: no sign-in**
3. **your project: exercise page**

1. Select a DataSet



Start

1. **header: DataSets**
2. **click the RNA-seq dataset**

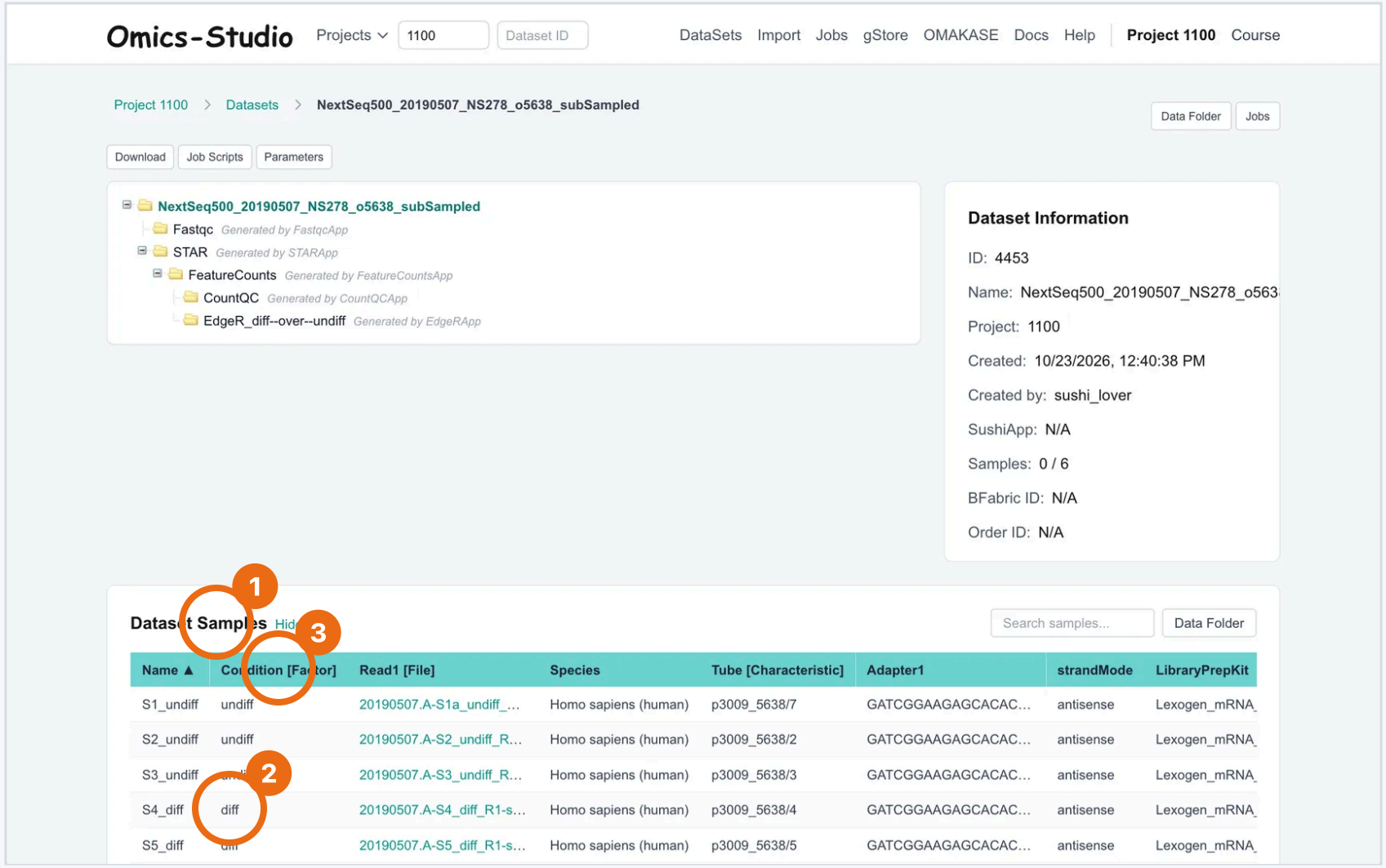
Note

tree: parent = input

child = output

link = one app run

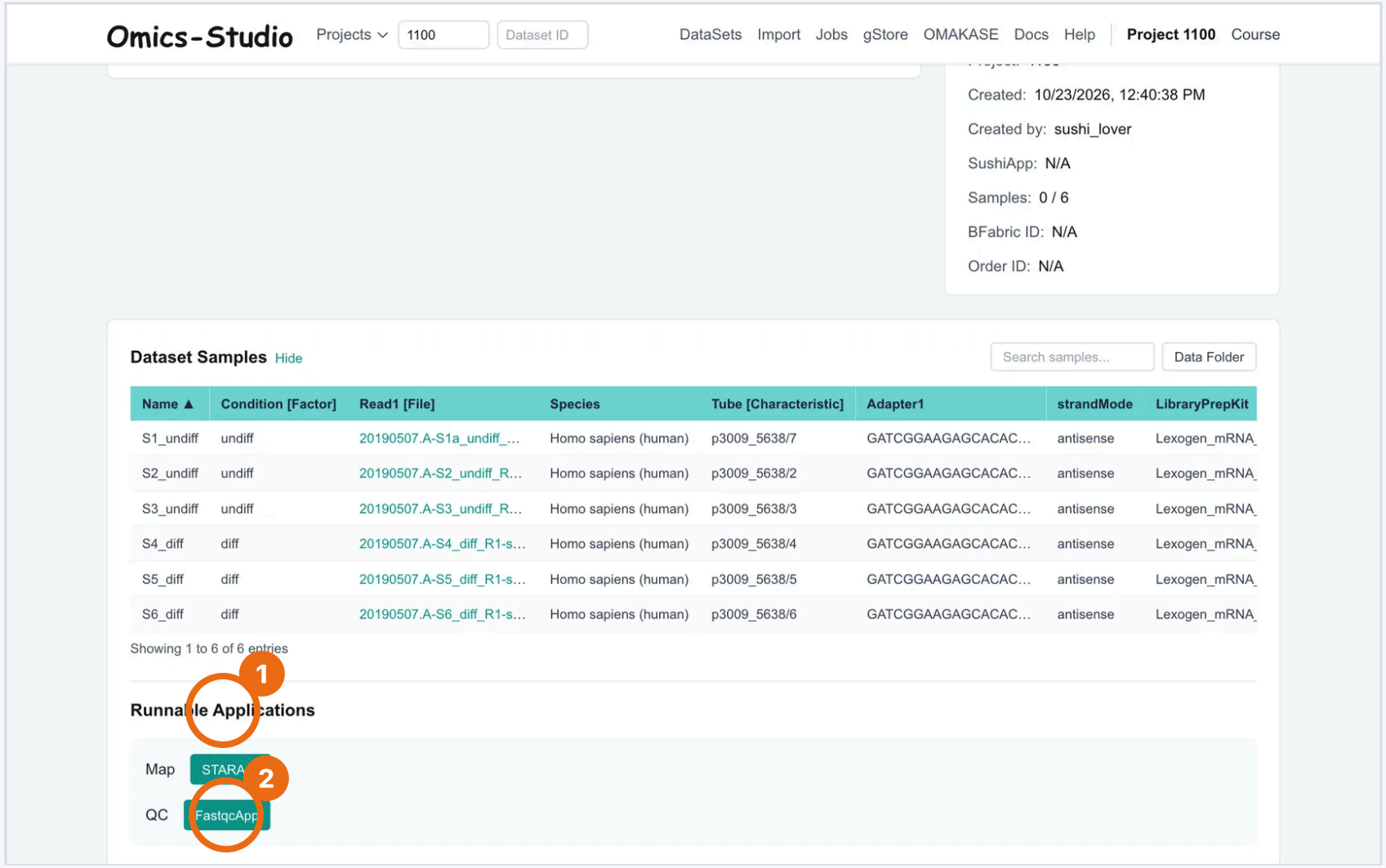
Check samples



Start

1. 6 samples
2. 3 undiff + 3 diff
3. Condition [Factor]: the groups

2. Select an application



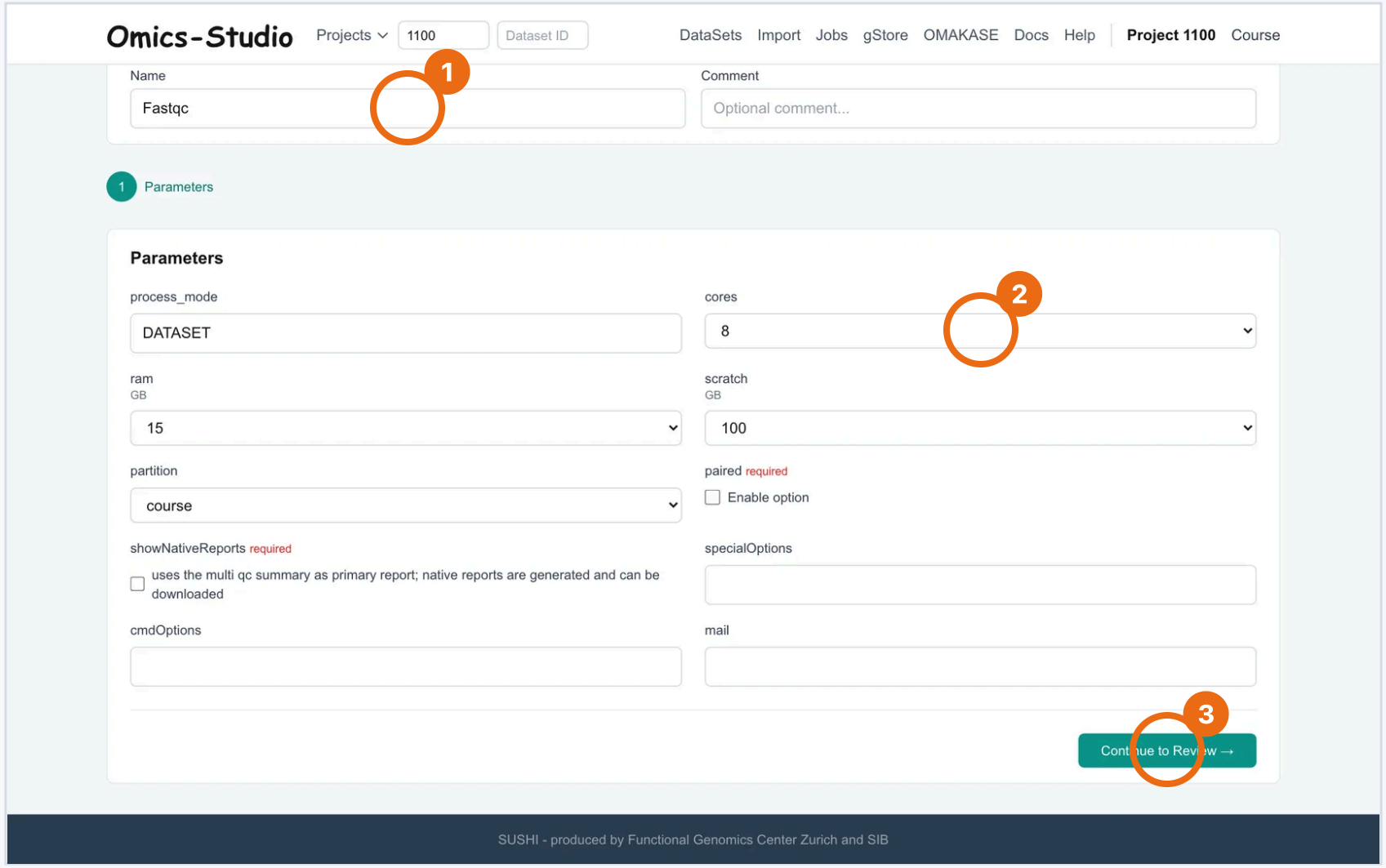
Exercise 1

1. **Runnable Applications**
2. **QC → FastqcApp**

Note

list follows the dataset type

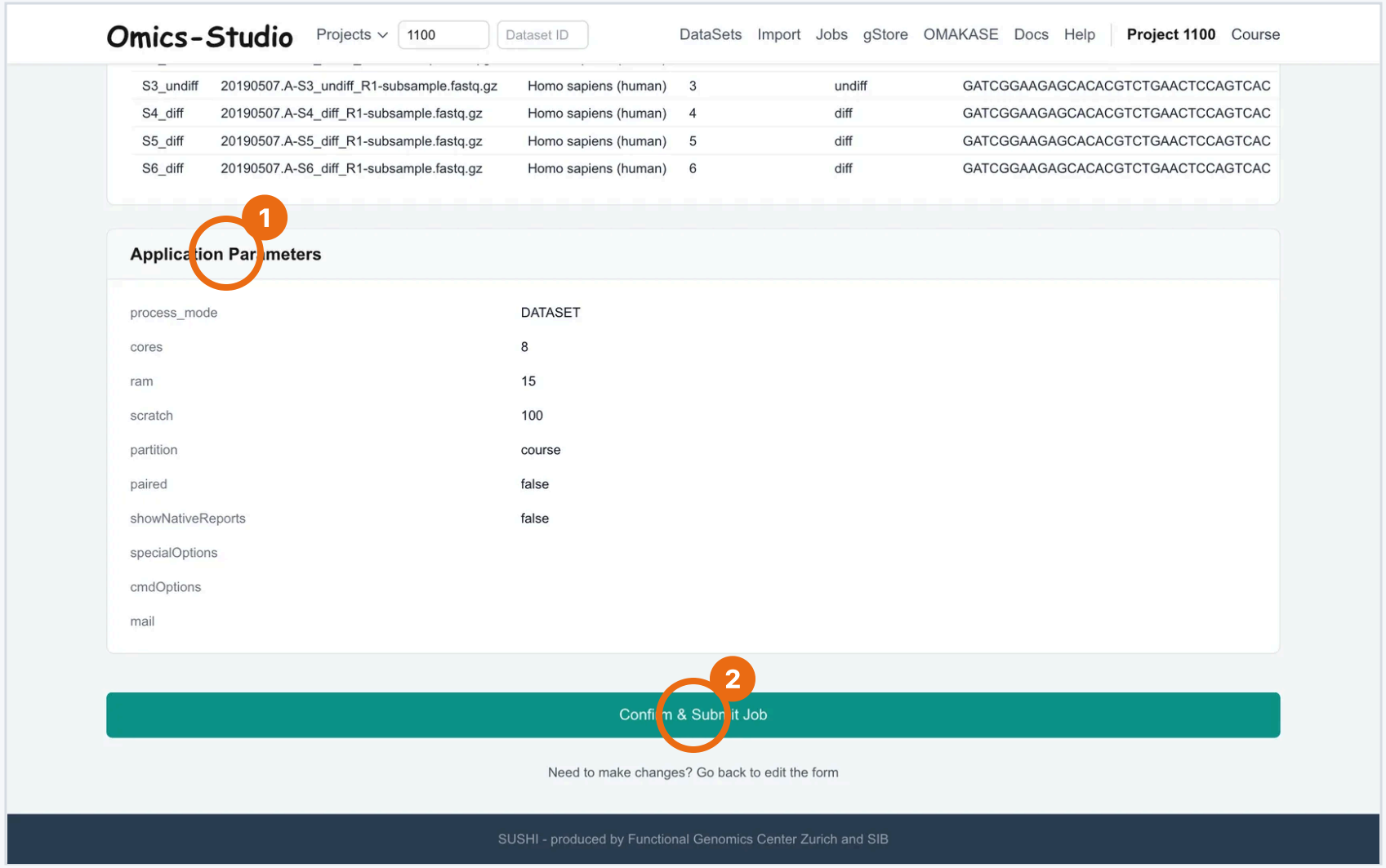
3. Set job parameters



Exercise 1

1. **Name: result folder**
2. **cores · RAM · scratch**
3. **defaults → Continue to Review**

4. Submit the job



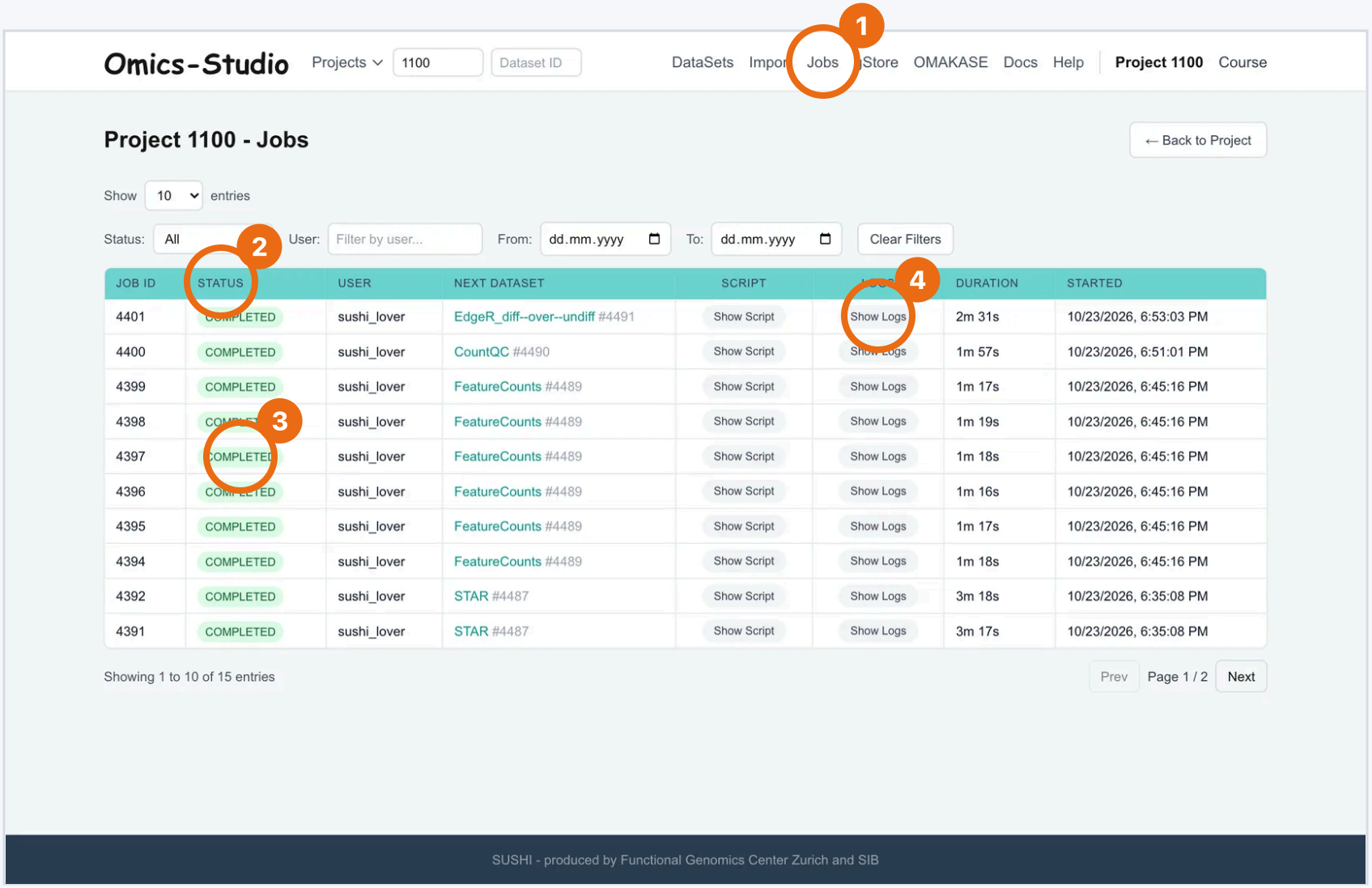
Exercise 1

1. **check parameters**
2. **Confirm & Submit Job**

Note

Omics-Studio writes a job script

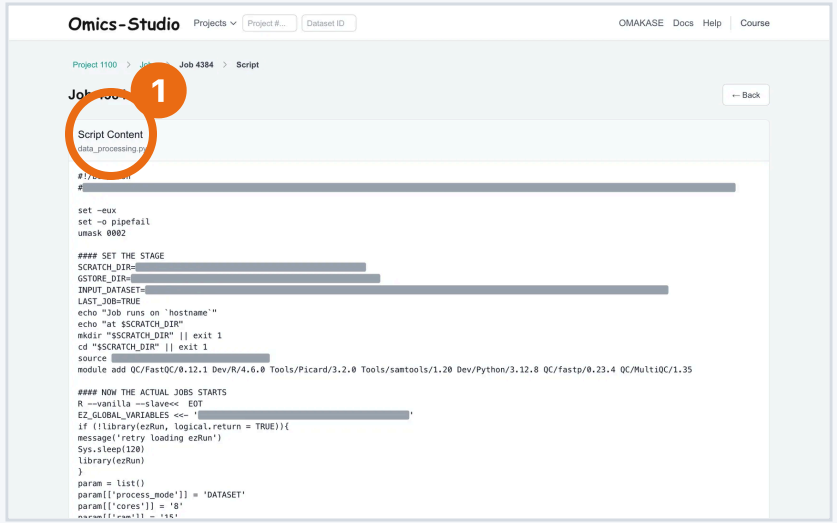
5. Check job status

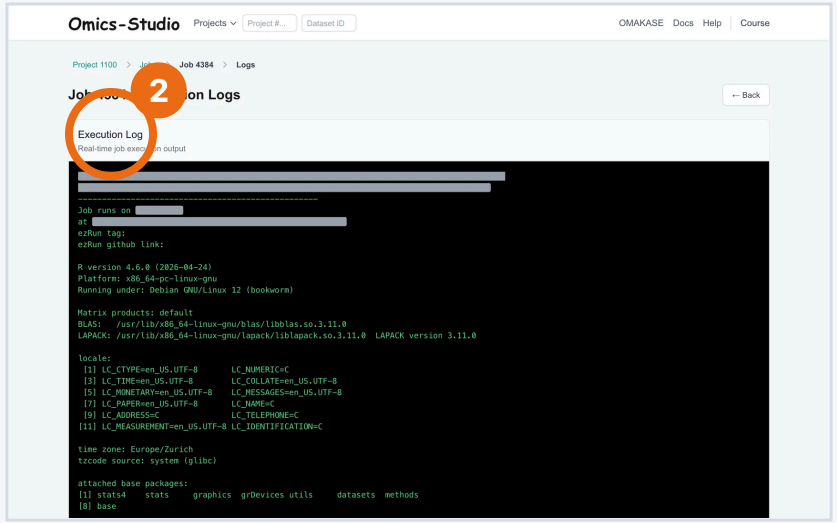


Exercise 1

1. **header: Jobs**
2. **STATUS: SUBMITTED, PENDING, RUNNING**
3. **COMPLETED: result ready**
4. **FAILED → Show Logs**

Job script and log





Exercise 1

1. **script:**
   what ran

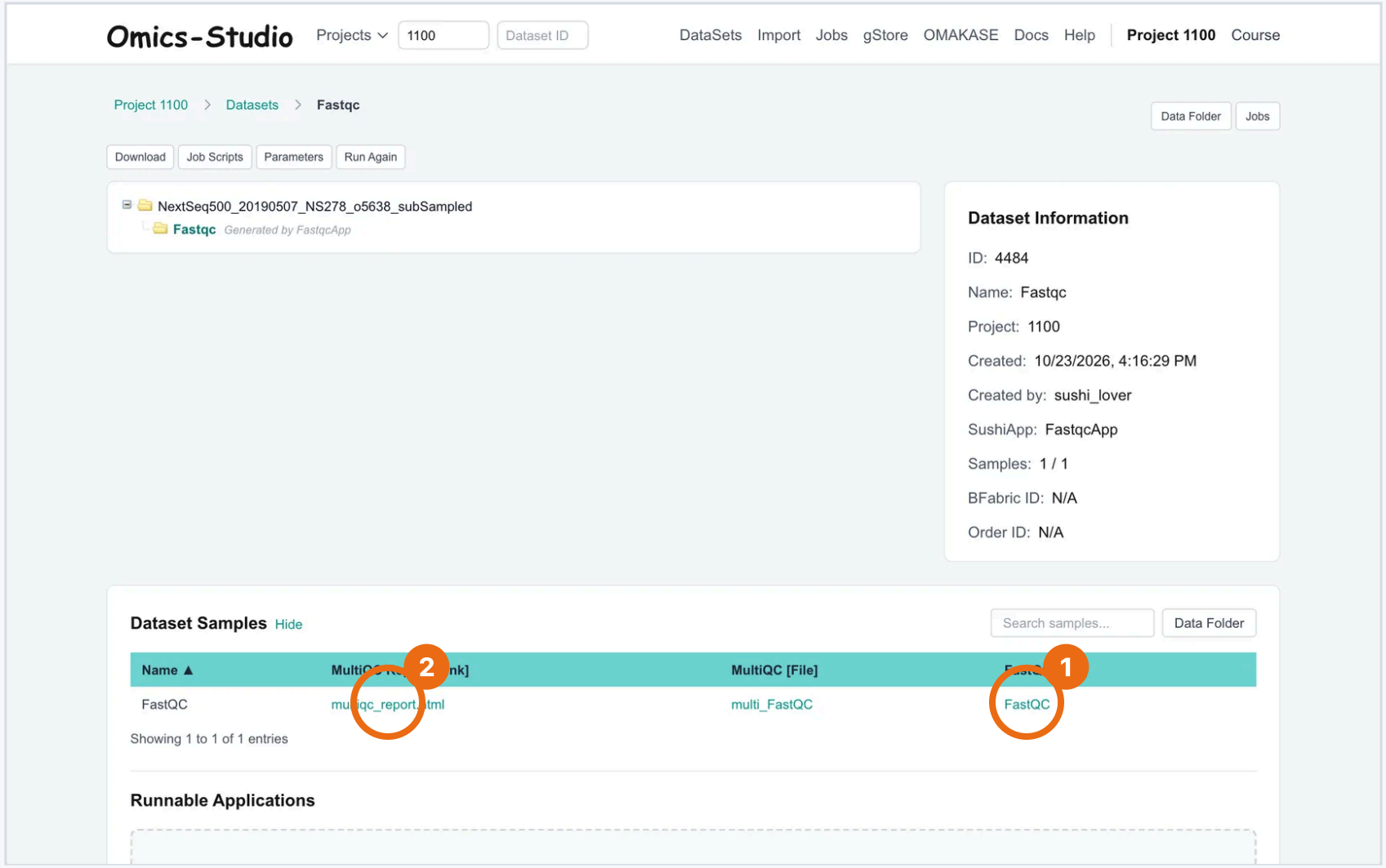
2. **log:**
   what it printed

Note

kept after the job:

you can redo it

FastQC result

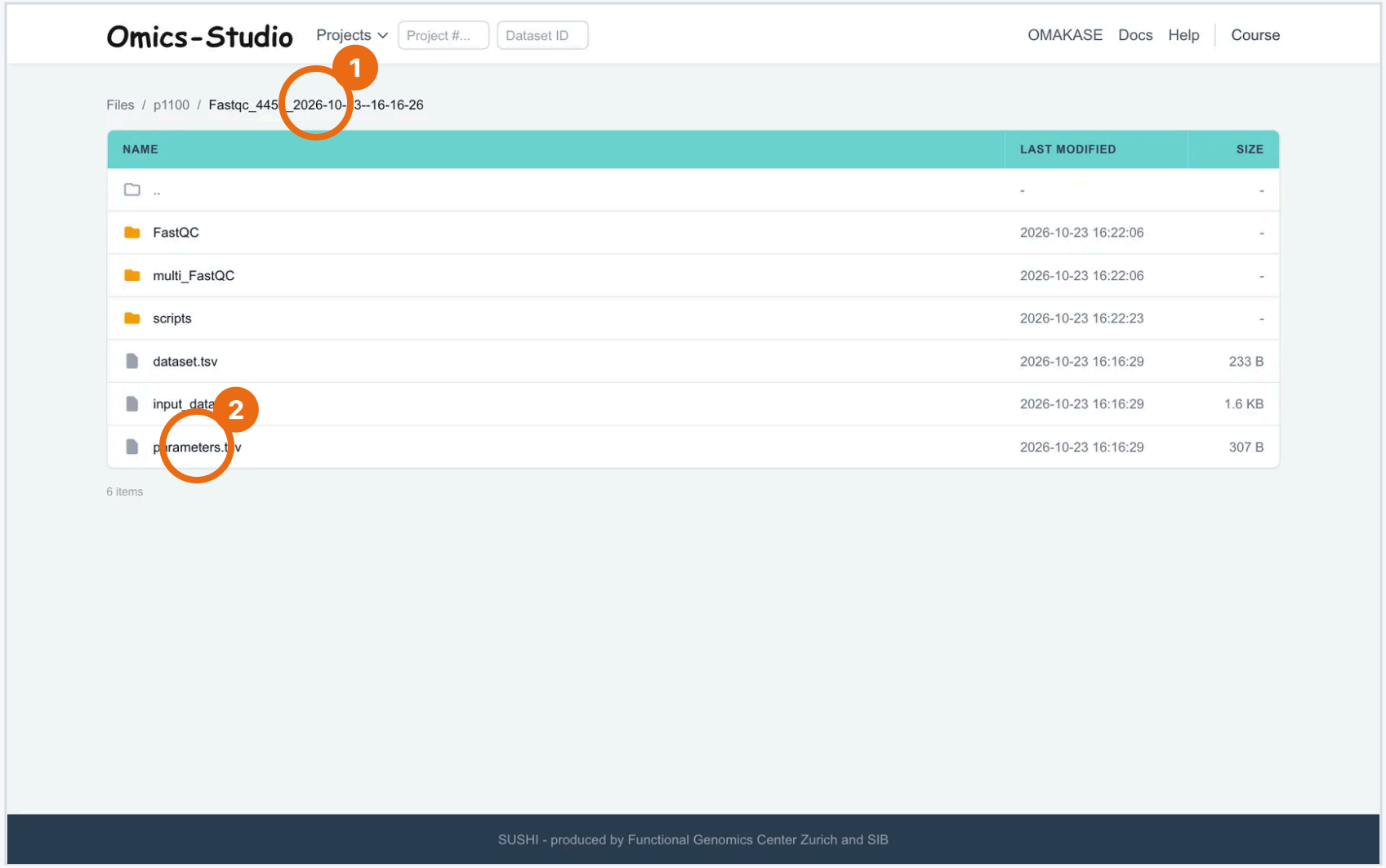


Exercise 1

1. **FastQC:**
   one report per sample

2. **MultiQC:**
   all samples together

Download a result



Exercise 1

1. **Data Folder → gStore**
2. **click a file: open or download**

Note

dataset table (.tsv)

≠ the data files

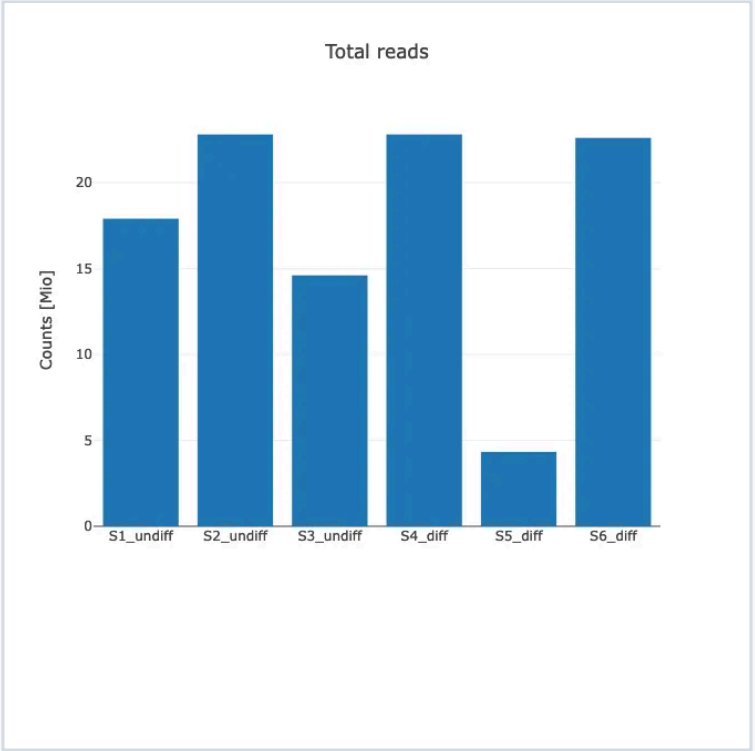
Exercise 1 · questions

Quantity

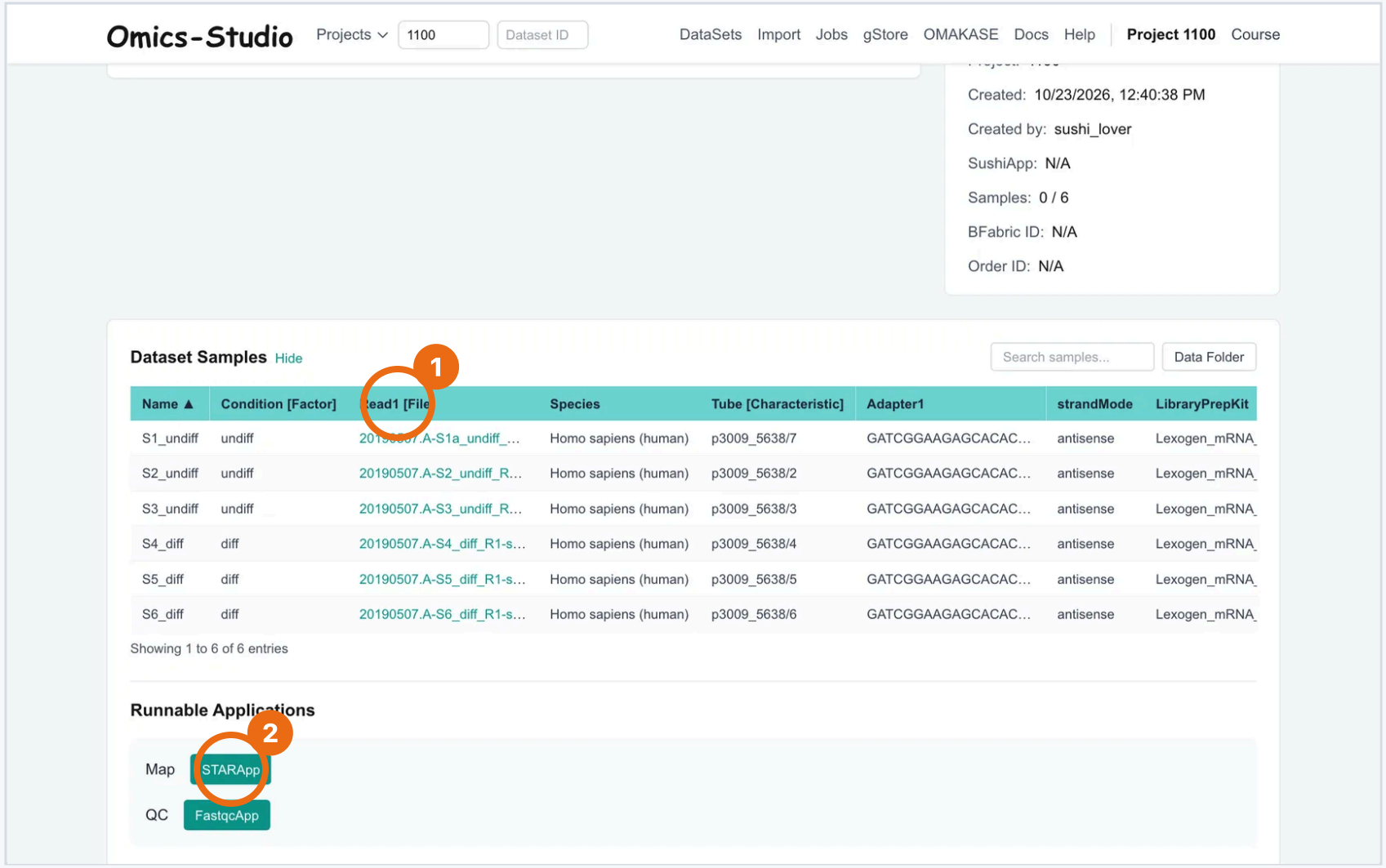
1. Which samples, if any, are overrepresented?

Quality

2. Base quality: differences across or within samples?
3. Adapter contamination in any sample? What might cause it?
4. A good preprocessing strategy for this data?



1. RNA-seq dataset → STARApp



Exercise 2

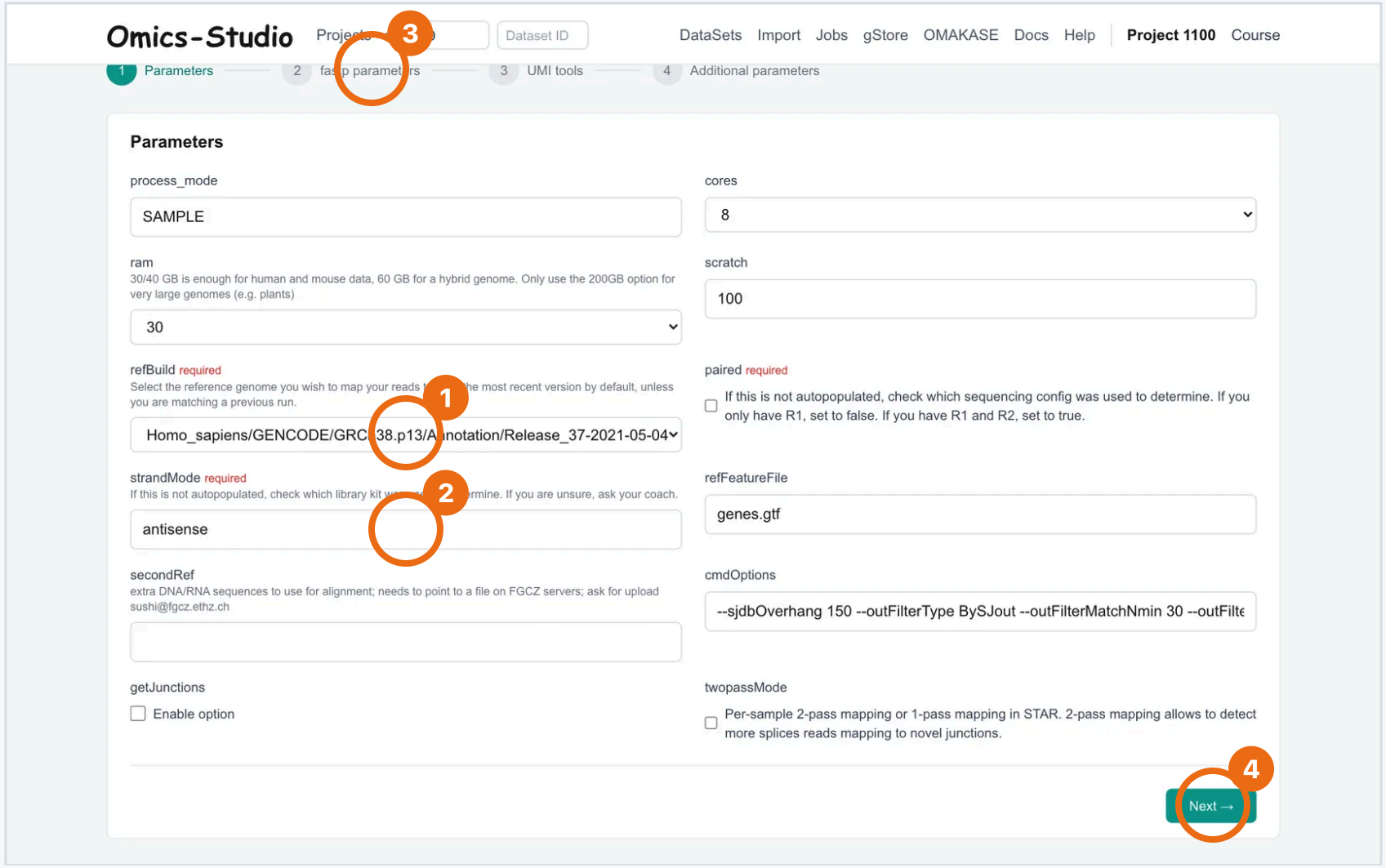
1. **back to RNA-seq dataset**
2. **Map → STARApp**

Note

FastQC result:

no apps (reports only)

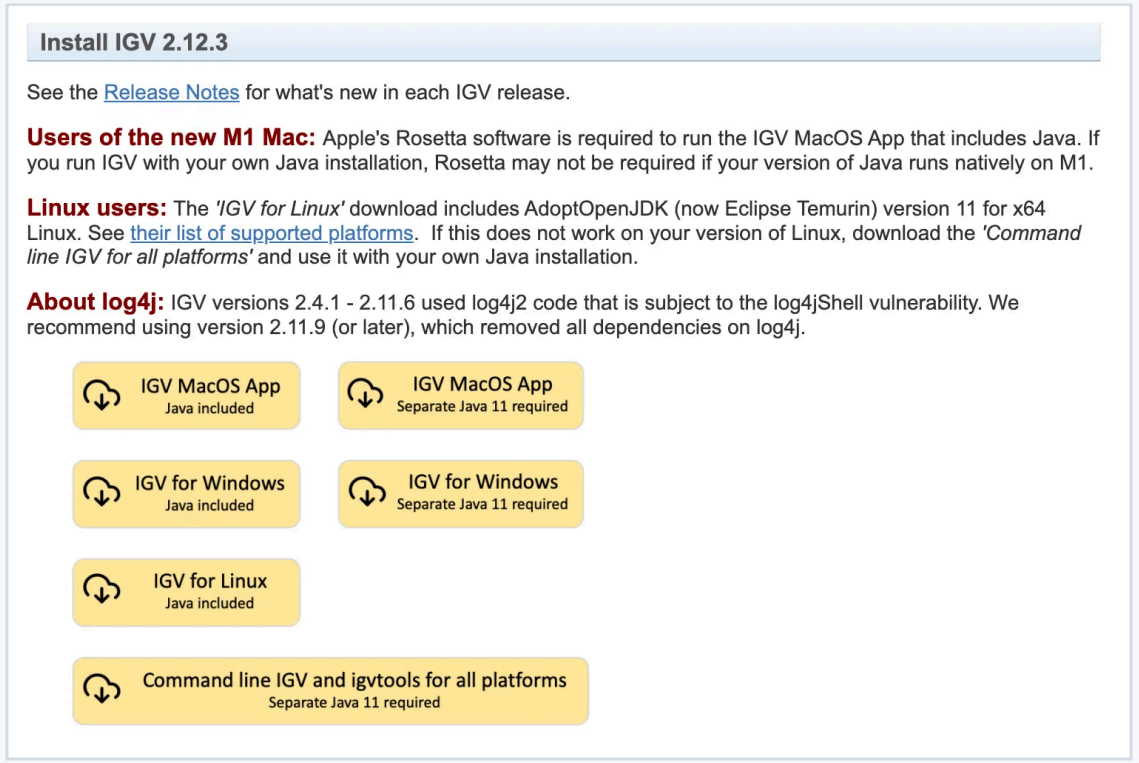
2. STAR parameters

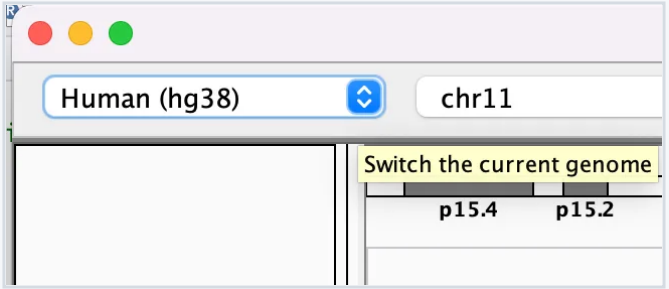


Exercise 2

1. **refBuild: GRCh38.p13 GENCODE Release 37**
2. **paired · strandMode: check them**
3. **fastp trim_front1 = 4**
4. **Next → … Review → Submit**

3. Get IGV





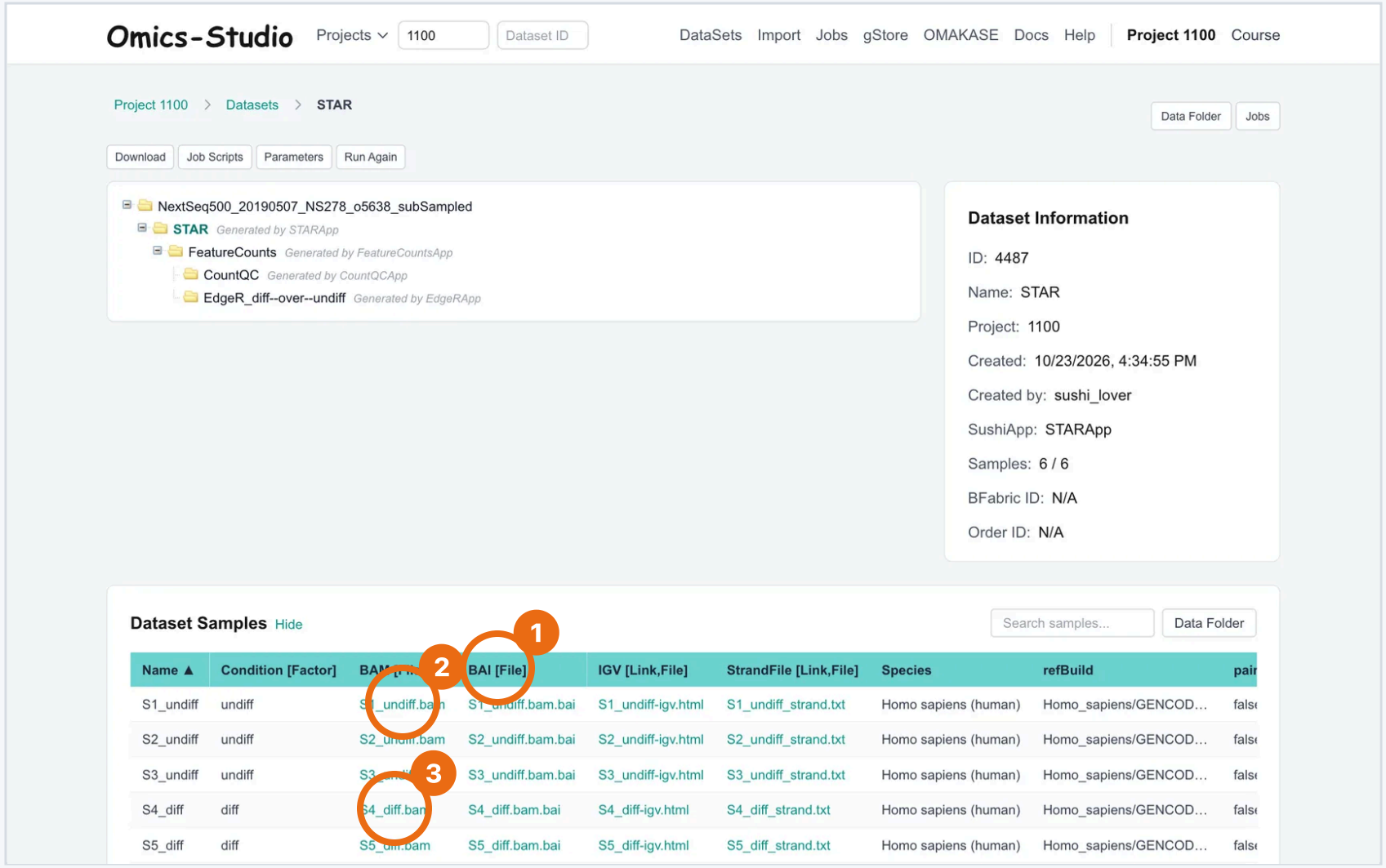
Exercise 2

1. **IGV: download for your laptop**
2. **genome: Human (hg38)**

Note

software.broadinstitute.org /software/igv/download

4. Copy the BAM links



Exercise 2

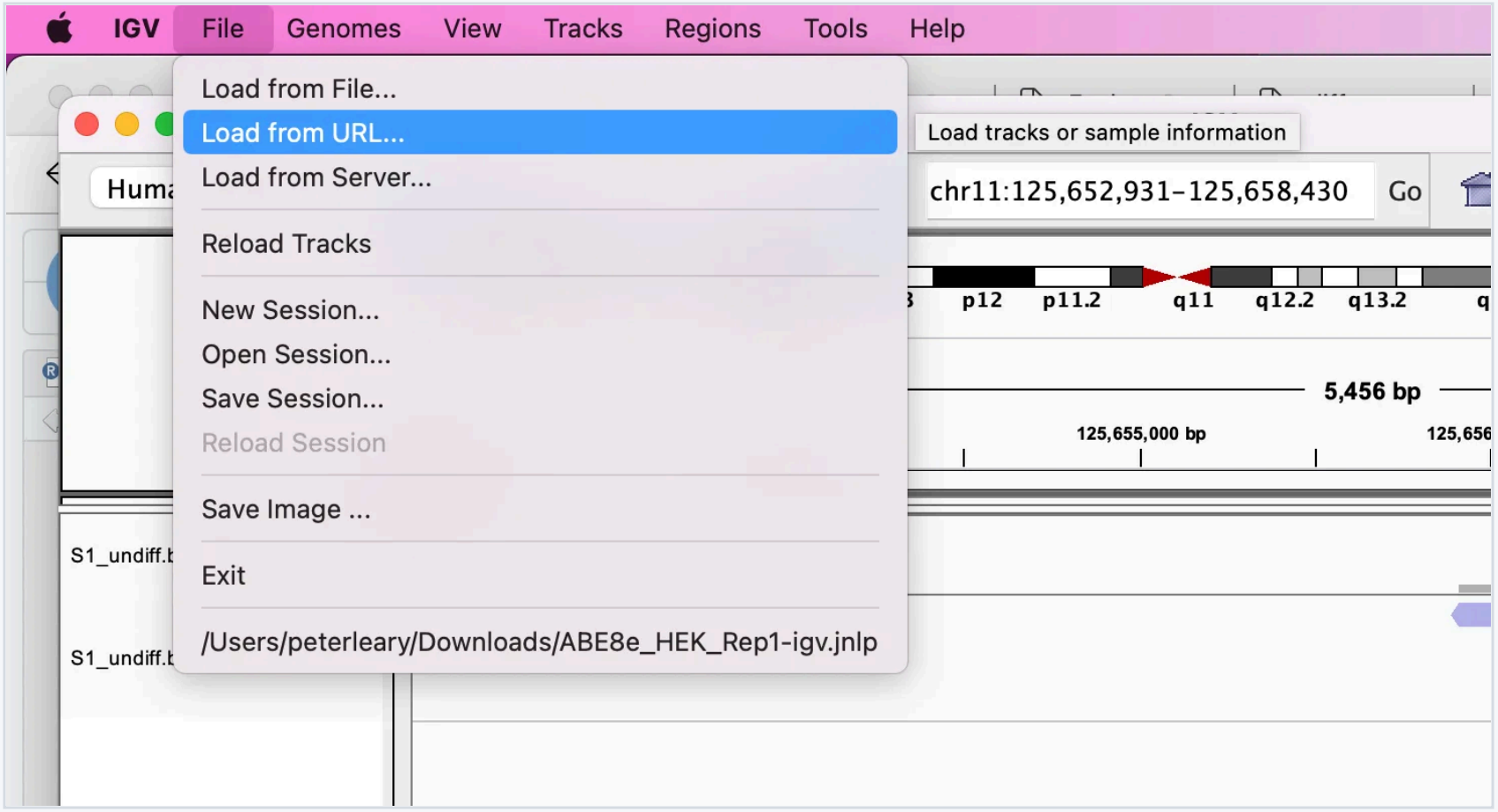
1. **STAR result: BAM · BAI columns**
2. **right-click → copy link**
3. **S1_undiff, S4_diff**

Note

not the IGV column:

asks for a sign-in

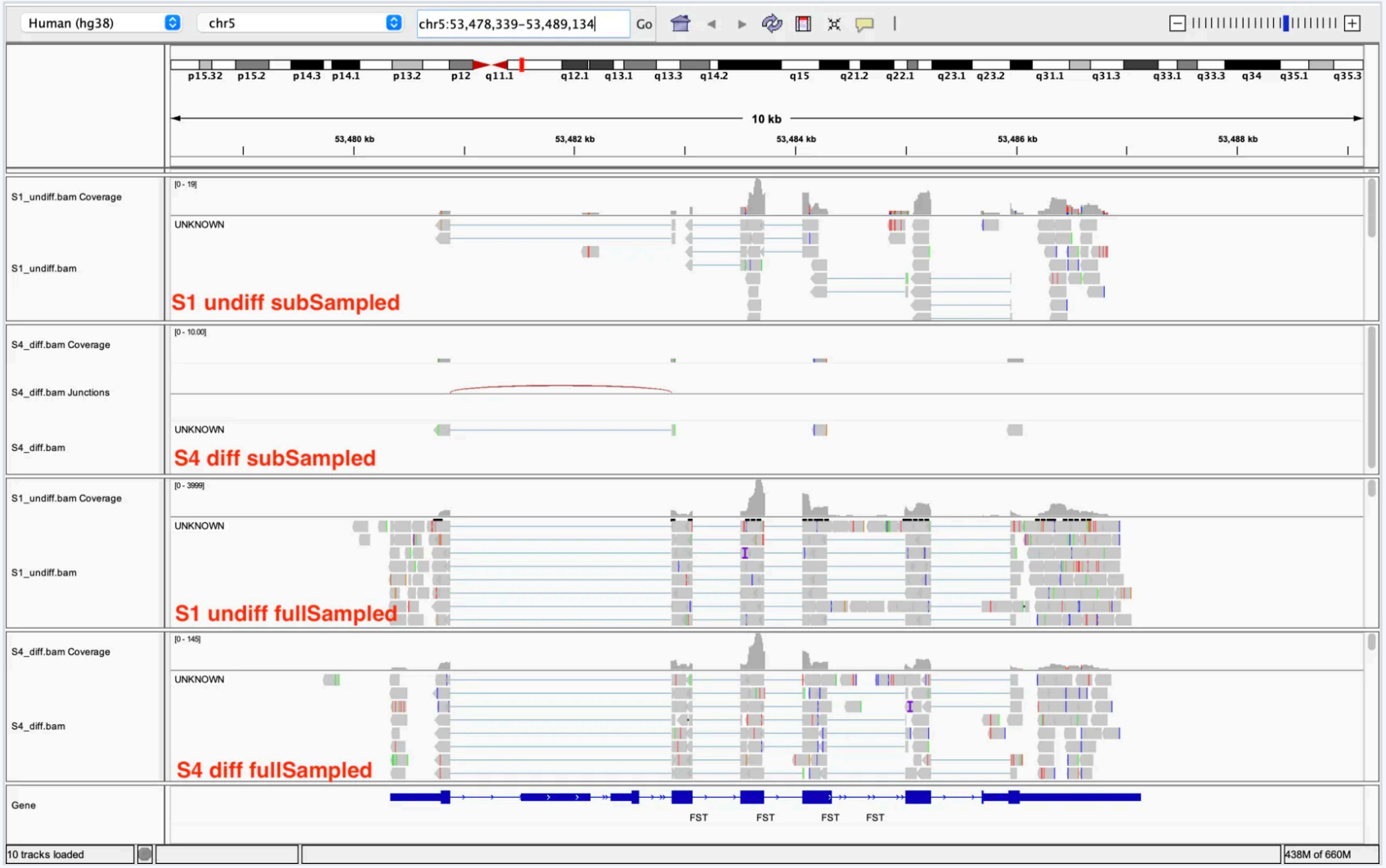
5. Load into IGV



Exercise 2

1. **File → Load from URL**
2. **paste BAM link + index link**

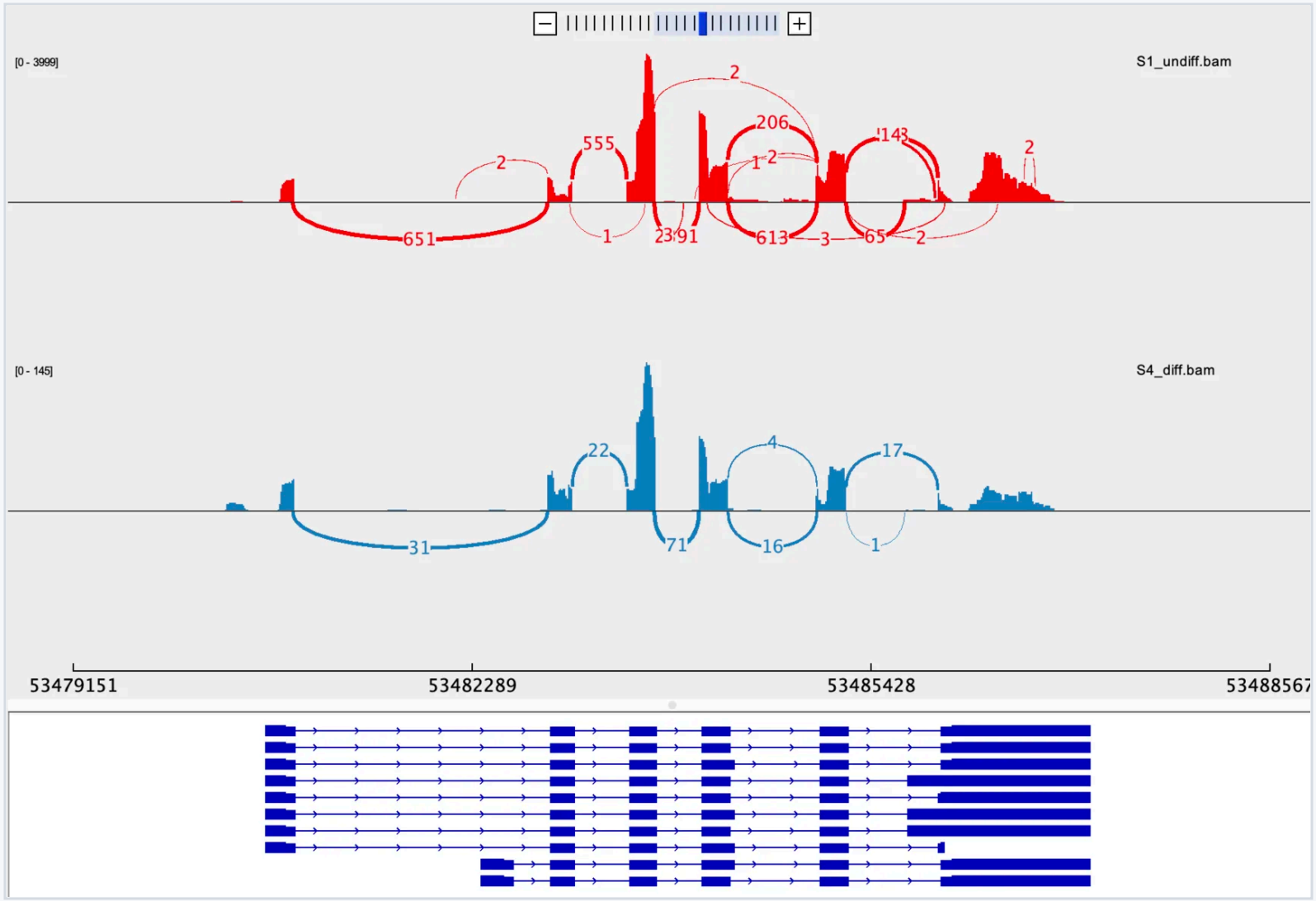
Reads in IGV



Exercise 2

1. **search: FST**
2. **coverage · reads · genes**
3. **split reads = splicing**

Sashimi plot



Exercise 2

1. right-click → Sashimi Plot
2. arcs: junctions, number of reads
3. colour = sample

Exercise 2 · questions

Exercise 2

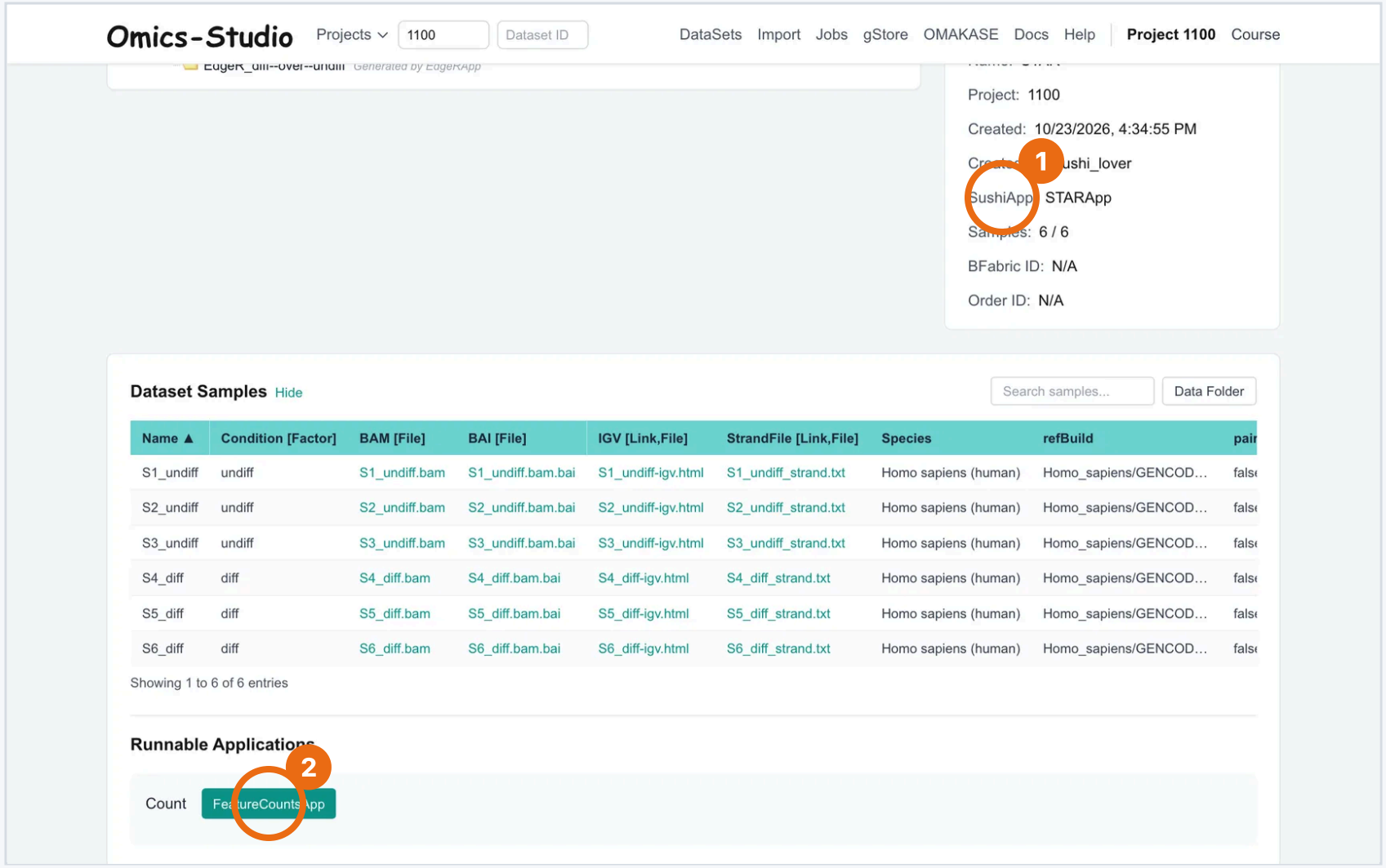
FST gene

1. Gene model: do the reads map to the right part of the gene?
2. Is the gene expressed in both conditions?
3. Coverage difference between the conditions?

Sashimi plot

4. How many isoforms are identified?
5. Which isoforms are present almost only in one sample? Which sample?

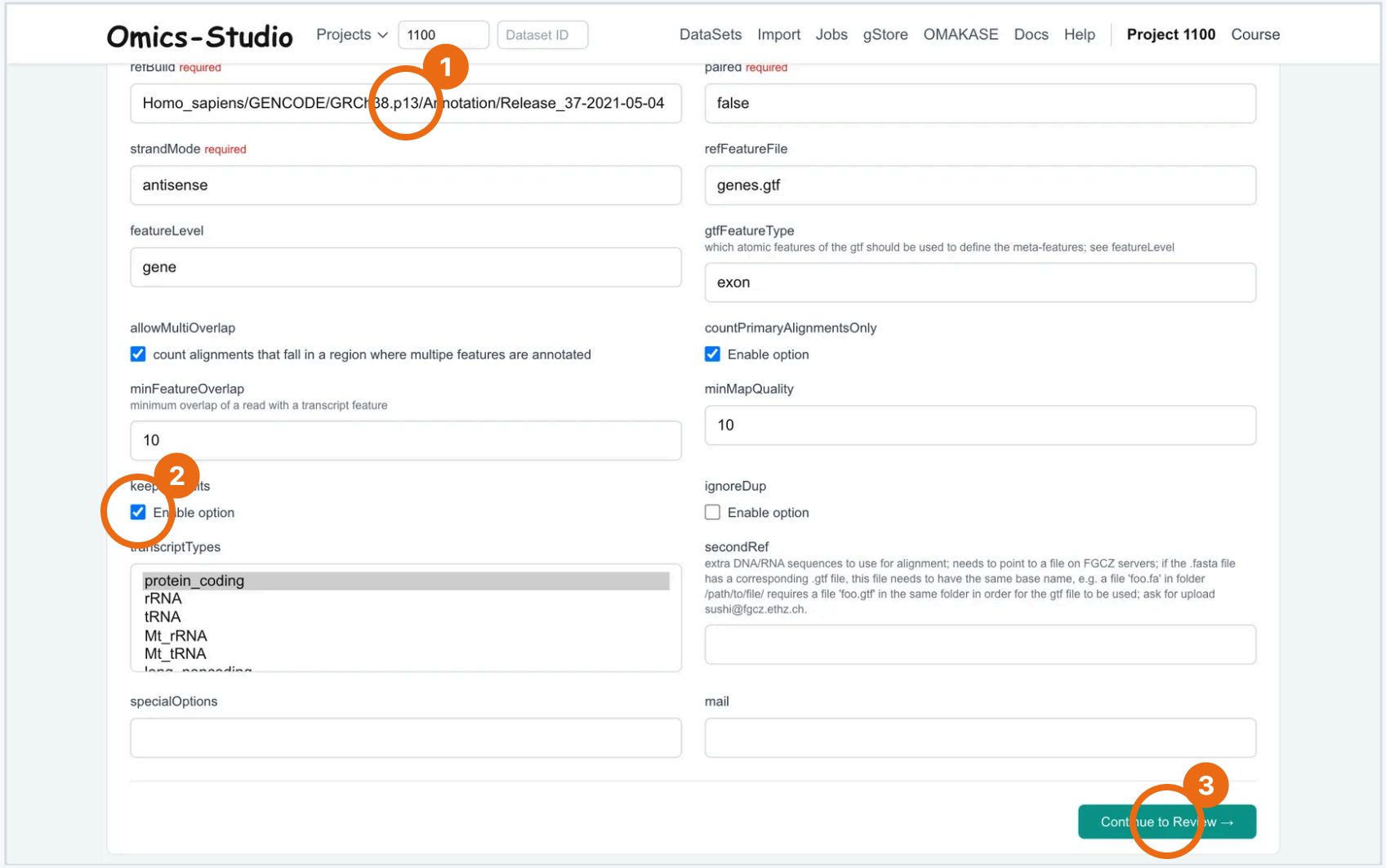
1. STAR result → FeatureCountsApp



Exercise 3

1. **STAR result dataset**
2. **Count → FeatureCountsApp**

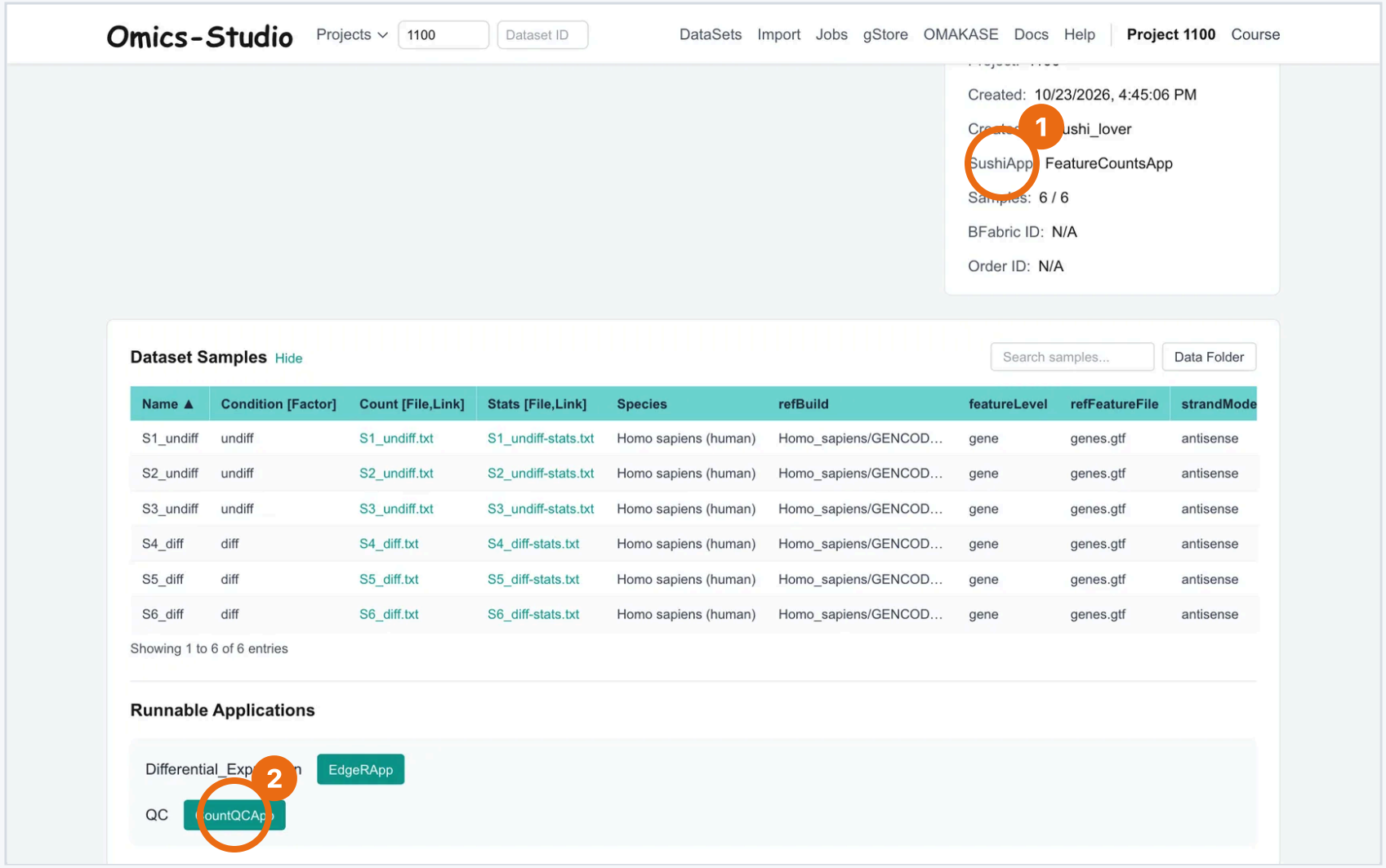
2. featureCounts parameters



Exercise 3

1. **refBuild · strandMode: from STAR**
2. **options: multi-hits, transcript types**
3. **defaults → Continue to Review**

3. featureCounts result → CountQCApp



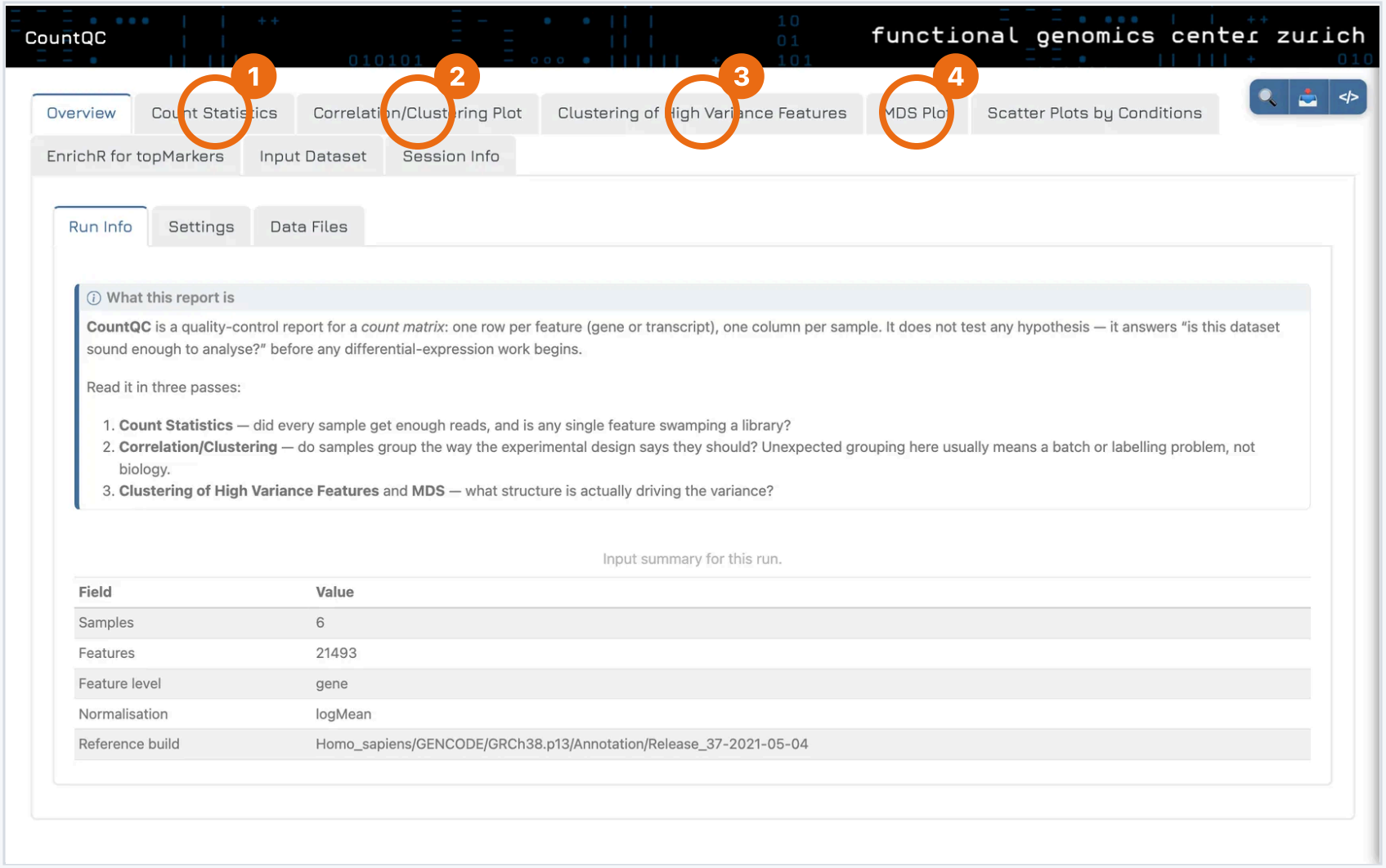
Exercise 3

1. **featureCounts result**
2. **QC → CountQCApp**

Note

edgeR starts from this dataset too

CountQC report



Exercise 3

1. **Count Statistics**
2. **Correlation / Clustering**
3. **High Variance Features**
4. **MDS Plot**

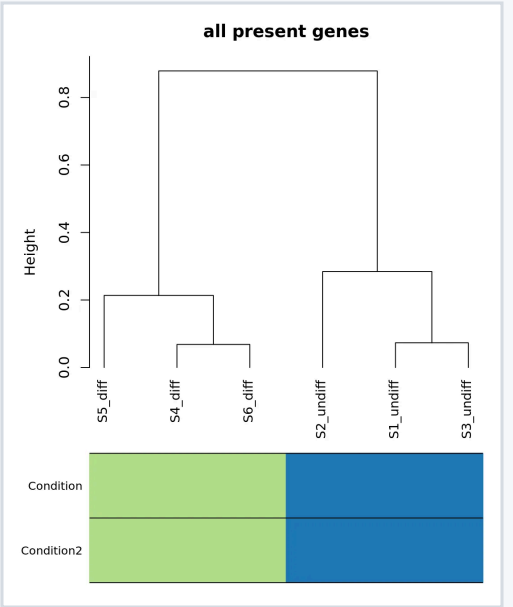
Exercise 3 · questions

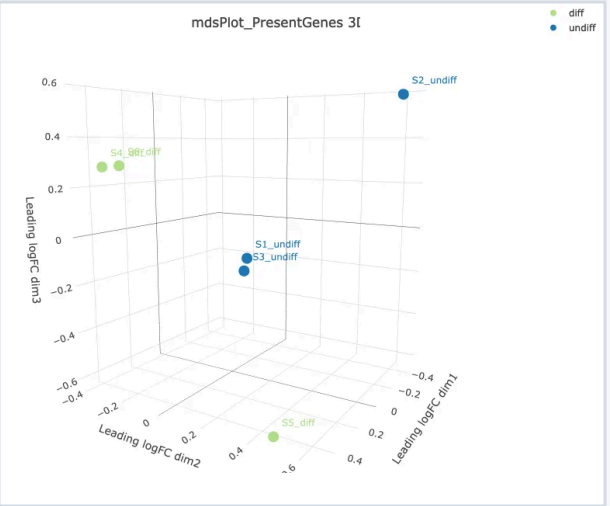
General QC

1. Reads in CountQC compared with FastQC?
2. Features above the threshold (in %)?
3. Underrepresented libraries: more reads, or exclude?

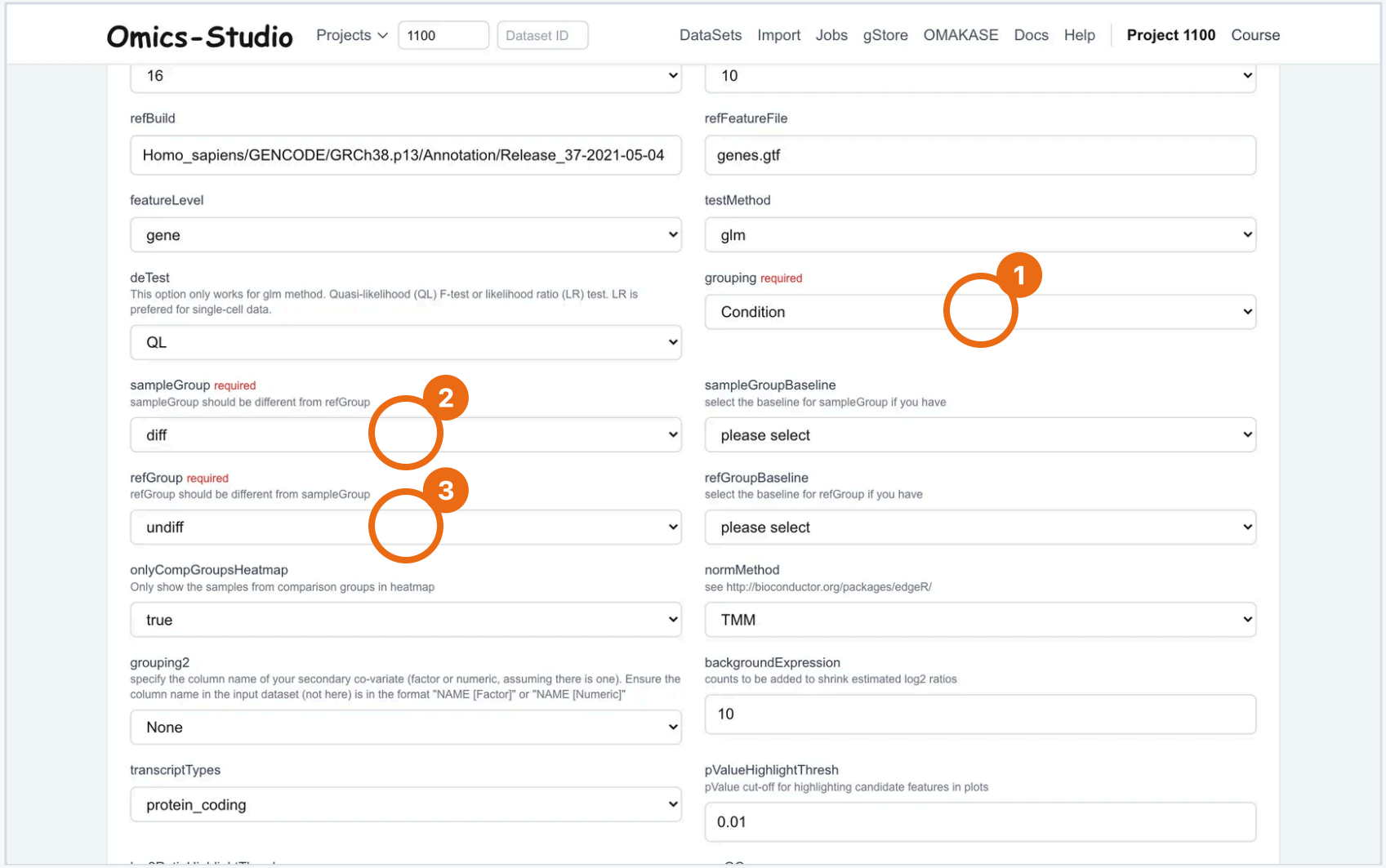
Clustering

4. How do the samples cluster?
5. Outliers? Keep or exclude?
6. Biologically relevant gene clusters?
7. MDS: how are the samples separated?





1. featureCounts result → EdgeRApp



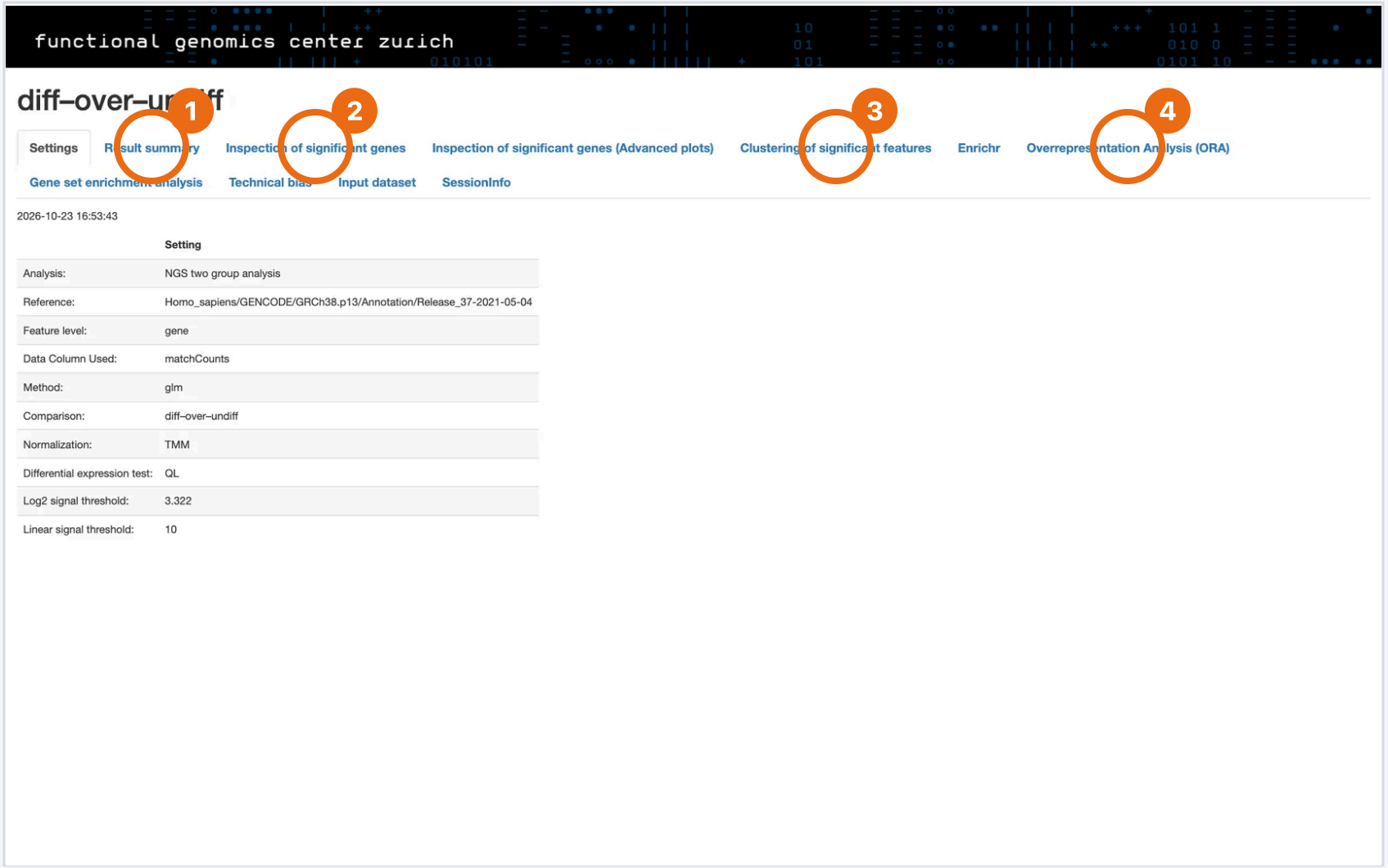
Exercise 4

1. **grouping: Condition**
2. **sampleGroup = diff**
3. **refGroup = undiff**
4. **rest: defaults**

Note

diff compared with undiff

edgeR report



Exercise 4

1. **result summary**
2. **significant genes: volcano**
3. **clustering: heat map**
4. **pathways: ORA · GSEA**

Note

correctBias = true:
rerun and compare

Exercise 4 · questions

Exercise 4

Results summary

1. Genes with p < 0.01 and log2 fold change ≥ ±2?
2. log2 fold change ≥ 1: how does the number of DE genes change with smaller p?

Significant genes

3. Volcano plot: how are the DE genes distributed?
4. The top candidate genes?

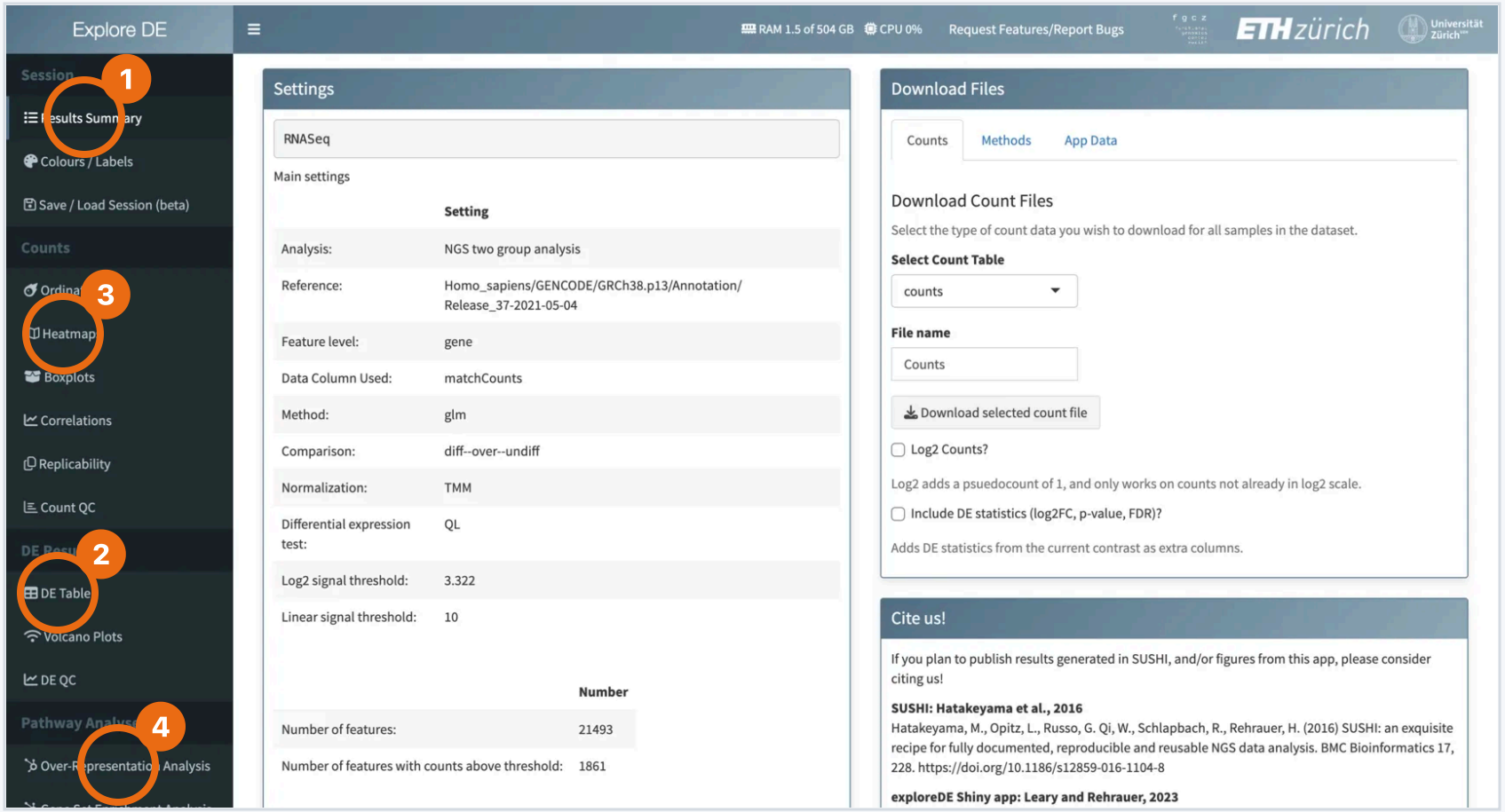
Clustering

5. DE genes in the heat map? Which thresholds?
6. Which condition has more up-regulated genes?

Pathways

7. GSEA: most enriched GO categories in the three ontologies?
8. ORA: which ontology has the most enriched categories?

Exercise 5 · at home: interactive report



Exercise 5

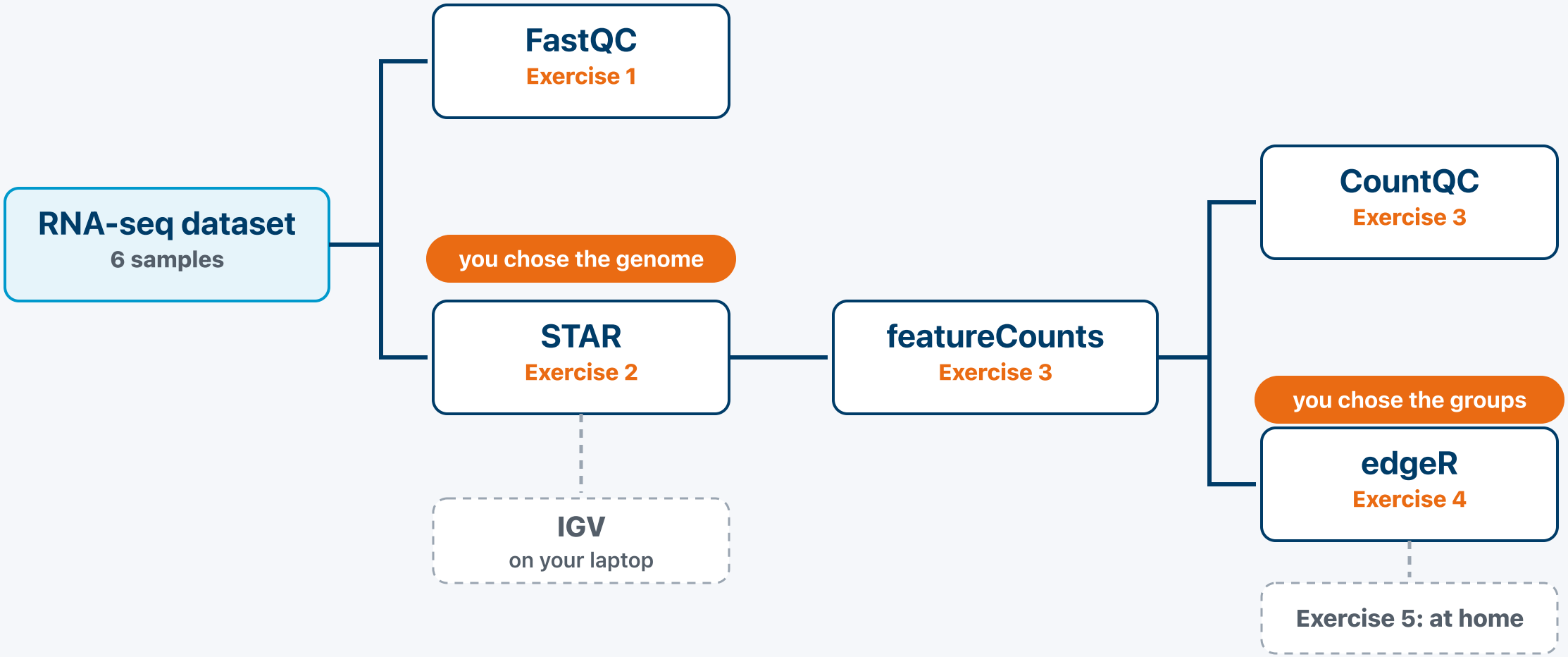
1. **Results Summary**
2. **DE Table · Volcano Plots**
3. **Heatmaps · Boxplots**
4. **Pathway Analyses**

Note

edgeR result →
Live Report (sign-in)

What you built: the whole chain, by hand

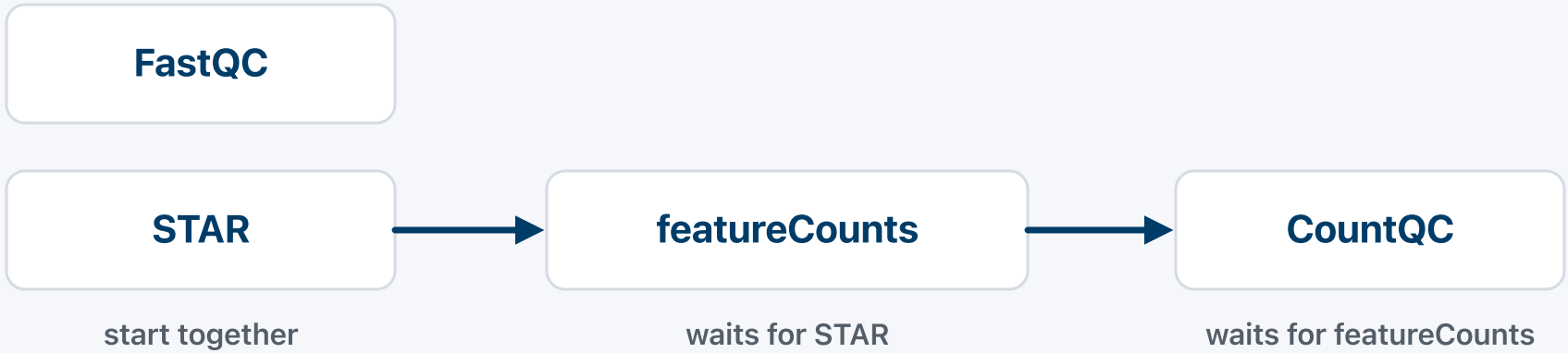
next: OMAKASE runs FastQC → CountQC from a recipe



Hands-on 2: OMAKASE runs the chain from a recipe



auto_recipe = Exercises 1–3 as one chain

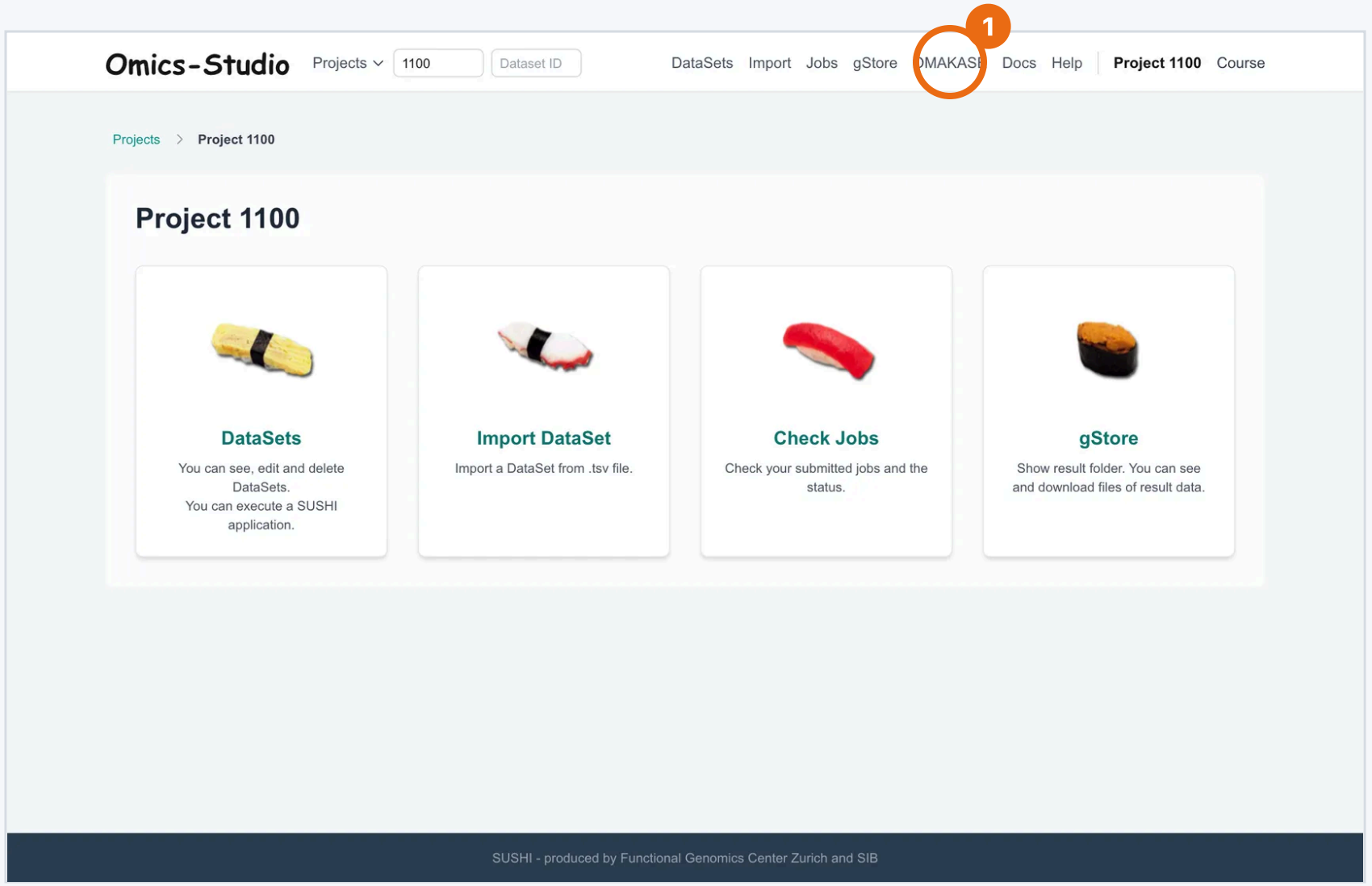


Where

fgcz-course2.bfabric.org
→ OMAKASE in the header
13.35–13.50

nothing runs until a person types their name, approves and presses Run chain

Open OMAKASE



Exercise 6

1. **header: OMAKASE**
2. **opens in a new tab**
3. **no sign-in: type your name**

Note

course assistant:
right-hand column,
28 Oct 13.00–15.00

1. Name, project, dataset → Propose



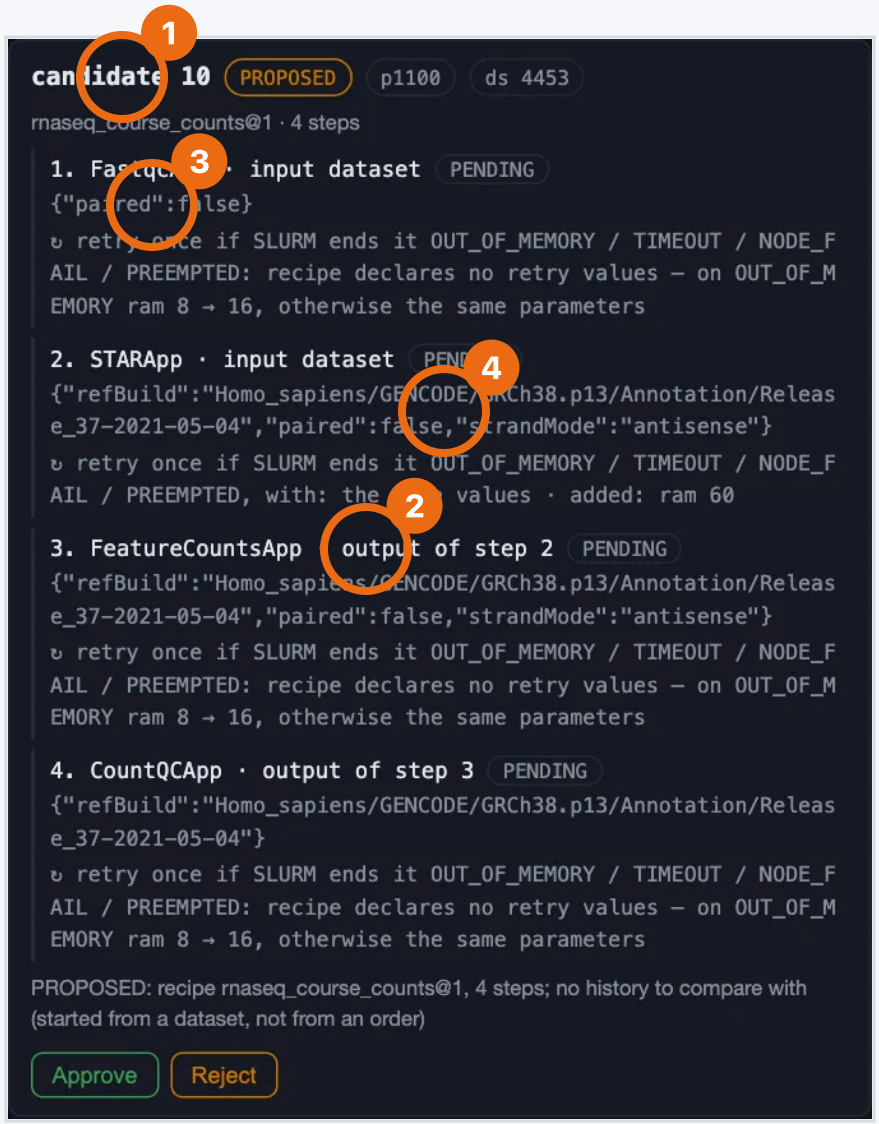
Exercise 6

1. **your name: required to approve**
2. **your project**
3. **dataset: the RNA-seq data**
4. **recipe: auto_recipe**
5. **Propose**

Note

Propose = a proposal: nothing runs yet

2. Read the proposal



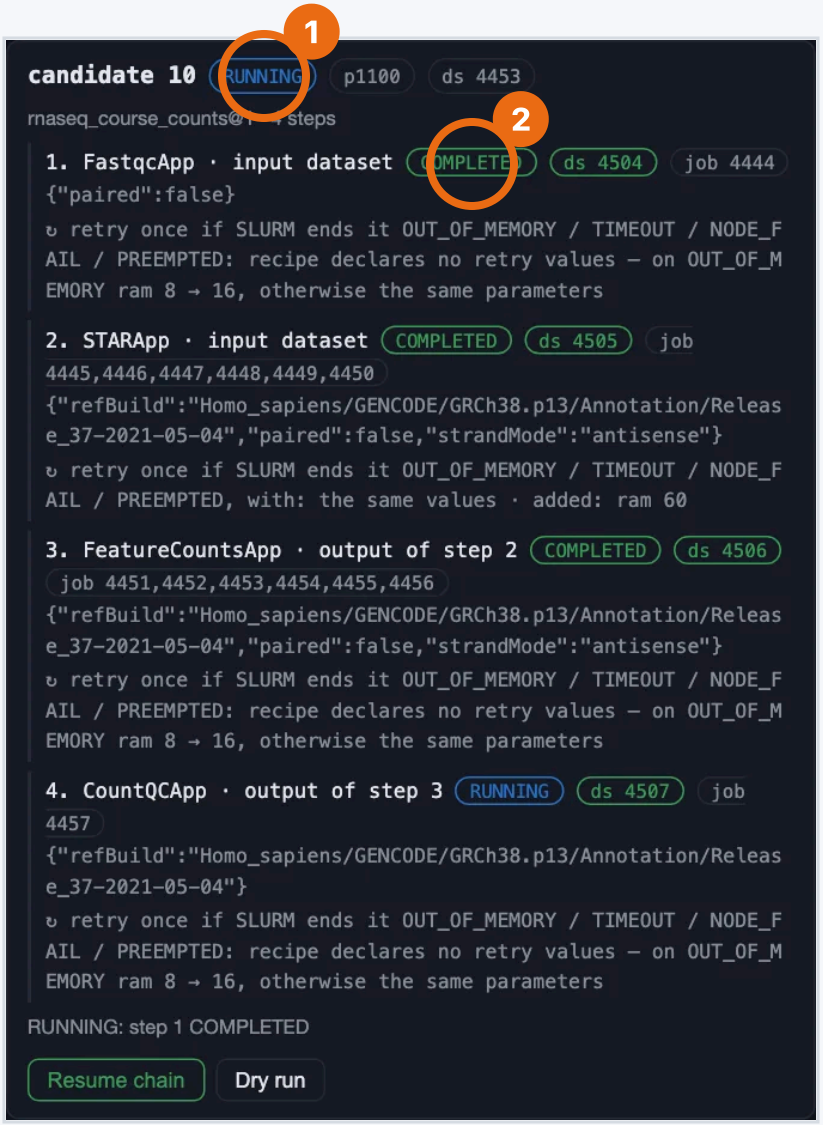
Exercise 6

1. **candidate number**
2. **steps:**
   which waits for which
3. **parameters**
4. **reference genome:**
   GRCh38.p13 Rel. 37

Note

nothing is submitted
before Approve

3. Approve → Run chain



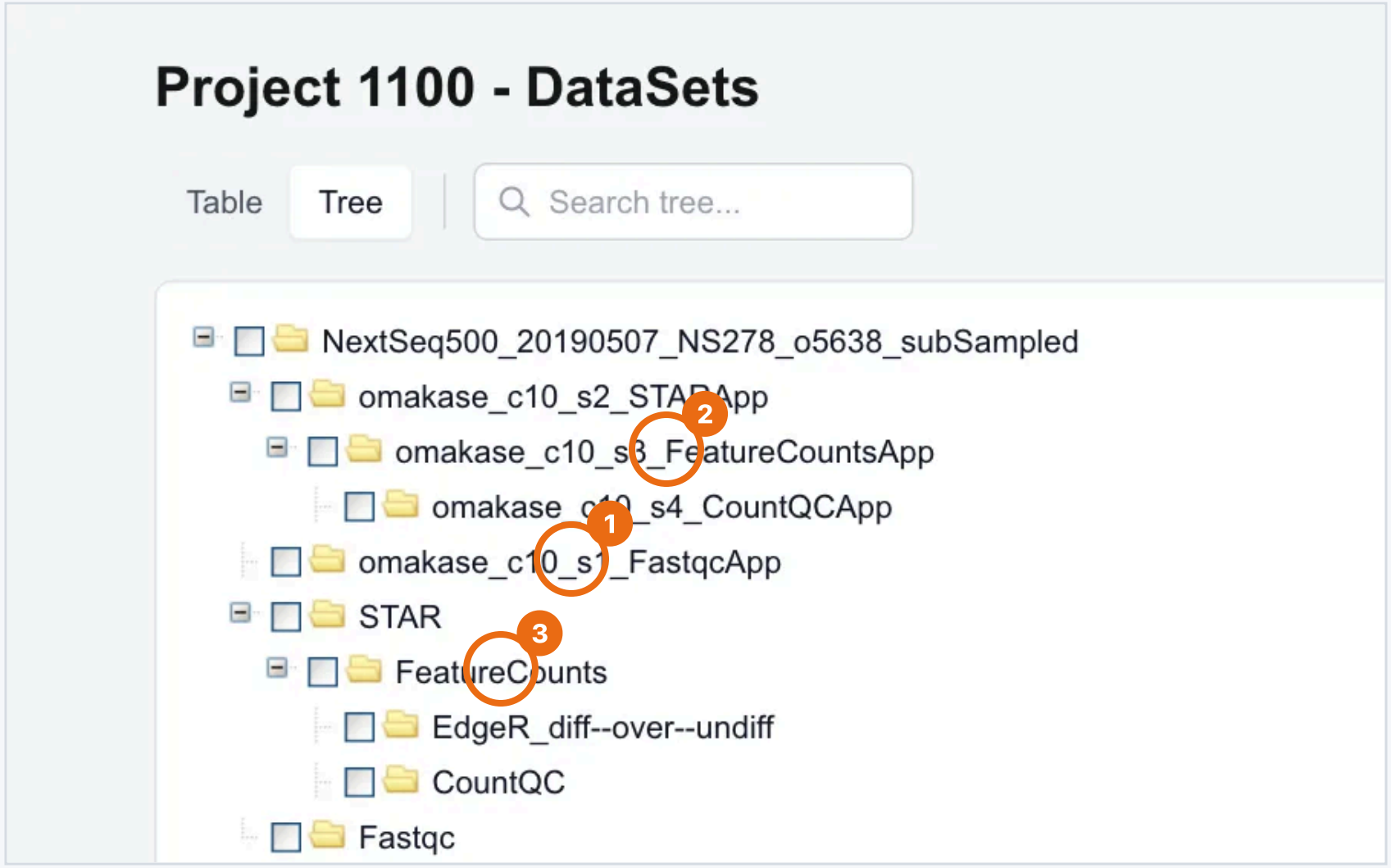
Exercise 6

1. **state: RUNNING**
2. **each step: COMPLETED, its dataset and jobs**
3. **about 8 min when the cluster is free**

Note

Run chain, not Dry run;
your typed name =
the approver

4. Results in Omics-Studio



Exercise 6

1. **DataSets: one per step**
2. **names: omakase_c..._s..._App**
3. **next to your Exercise 1–3 results**

Run it again: Reset this project



Exercise 6

1. **your name**
2. **Reset this project: propose again**

Note

only your own project

Exercise 6 · questions

Exercise 6

Rules decide

1. OMAKASE chose the recipe by a rule. Why does it still ask a person before anything runs?
2. Which reference genome is in the proposal? The one from Exercise 2? Why does it matter?

The chain

3. Which jobs started together? Why does featureCounts wait for STAR but not for FastQC?
4. If STAR had failed, what should happen to featureCounts?

Humans release

5. Where do you see who approved? With no sign-in, what does a typed name mean?
6. The recipe stops at CountQC. Which choice in Exercise 4 is hard to automate, and why?
7. By hand or by recipe: what does each record about how the result was made?

Same tree, two ways to grow it

By hand · Exercises 1–4

you chose each app
and its settings
record: datasets + job scripts

By recipe · Exercise 6

a recipe fixed the chain
and the settings; a person approved
record: the same + who released it

Rules decide, AI assists, humans release.

next: wrap-up